



NOVA SCHOOL OF
SCIENCE & TECHNOLOGY

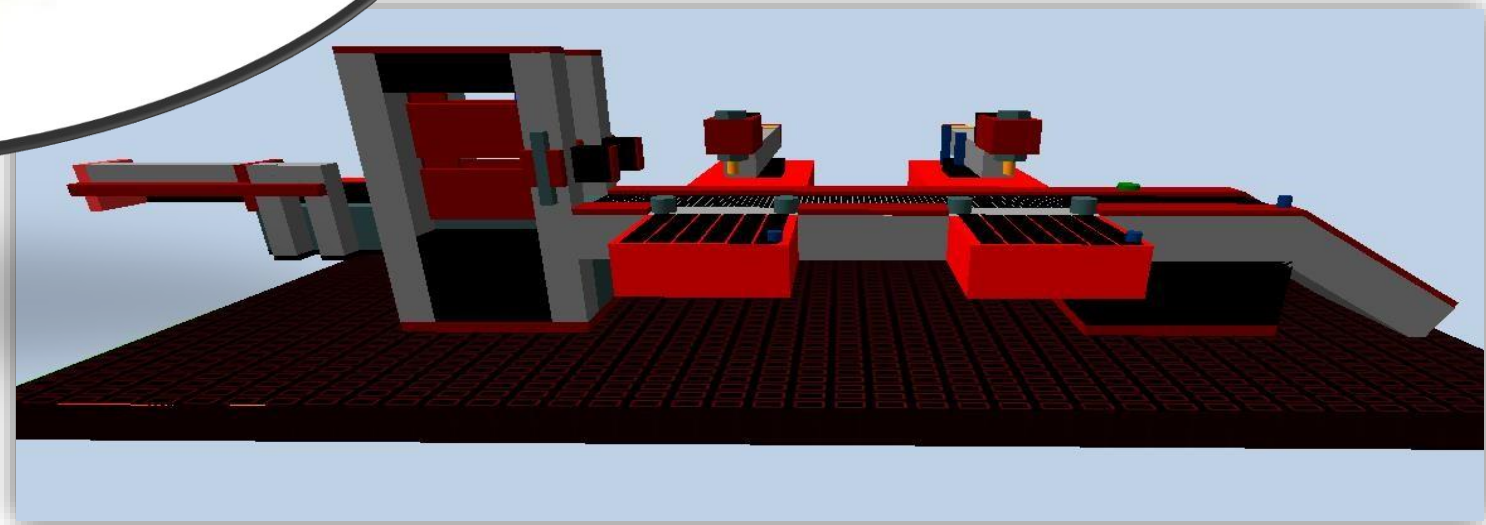
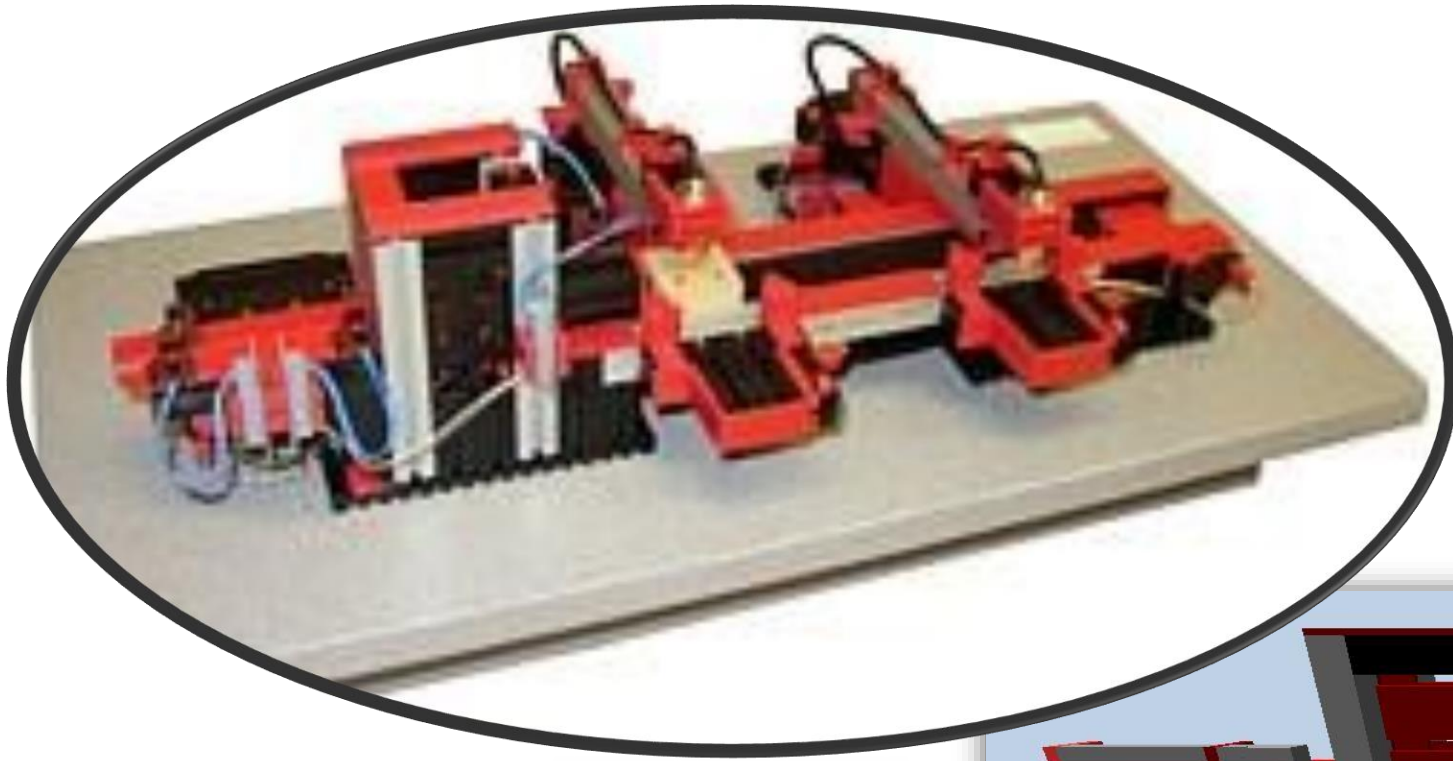
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LAB #6

Java SDK and Netbeans installation

LABWORK 2: interaction with Physical System

2022 - 2023



□ 5 classes

Class 1	27/28/31 October 2022	Lab#6	SW Installation & Interaction with “physical” system. Small Java tutorial and low-level functionalities implementation.
Class 2	03/04/07 November 2022	Lab#7	JNI development for interacting with the physical system from Java. Elaboration of both low-level and high-level routines for LabWork2.
Class 3	10/11/14 November 2022	Lab#8	Introduction to Threads and synchronization and communication mechanisms. Development of the necessary threads for modeling LabWork2 concurrent behavior.
Class 4	17/18/21 November 2022	Lab#9	Fulfillment of each functional requirement towards the solution for LabWork2.
Class 5	24/25/28 November 2022	Lab#10	Conclusion of LabWork2.



Deadline for Delivery: 03-12-2022

PART 1:

Software installation & configuration

- ❑ Software installation
 - [Java SDK Installation](#)
 - [Small Java tutorial](#)
 - [Netbeans Installation](#)

PART 2:

Interaction with physical system

- ❑ [Introducing the kit “Splitter Conveyor”](#)
- ❑ [Creation of Visual studio console application project](#)
- ❑ [Development of low-level functionality](#)
- ❑ [Testing each kit movement](#)

PART 1

Software installation & configuration

- Remind
- **If you have got a working Java SDK 1.8 and Netbeans, then you can skip these steps and proceed to [Java Examples...](#)**



Java SDK Installation



Download JavaSDK: <https://www.oracle.com/java/technologies/javase/javase8u211-later-archive-downloads.html>

Java SE 8 Archive Downloads (JDK 8u211 and later)

Go to the [Oracle Java Archive](#) page.

The JDK is a development environment for building applications using the Java programming language.

The JDK includes tools useful for developing and testing programs written in the Java programming language and running on the Java™ platform.

Important Oracle JDK License Update

The Oracle JDK License has changed for releases starting April 16, 2019.

The new [Oracle Technology Network License Agreement for Oracle Java SE](#) is substantially different from prior Oracle JDK licenses. The new license permits certain uses, such as personal use and development use, at no cost -- but other uses authorized under prior Oracle JDK licenses may no longer be available. Please review the terms carefully before downloading and using this

prod	Solaris SPARC 64-bit (SVR4 package)	133.66 MB	 jdk-8u301-solaris-sparcv9.tar.Z
Com	Solaris SPARC 64-bit Compressed Archive	94.8 MB	 jdk-8u301-solaris-sparcv9.tar.gz
Orac	Solaris x64 (SVR4 package)	134.42 MB	 jdk-8u301-solaris-x64.tar.Z
	Solaris x64 Compressed Archive	92.66 MB	 jdk-8u301-solaris-x64.tar.gz
	Windows x86 Installer	156.45 MB	 jdk-8u301-windows-i586.exe
	Windows x64 Installer	169.46 MB	 jdk-8u301-windows-x64.exe

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Java SE 8

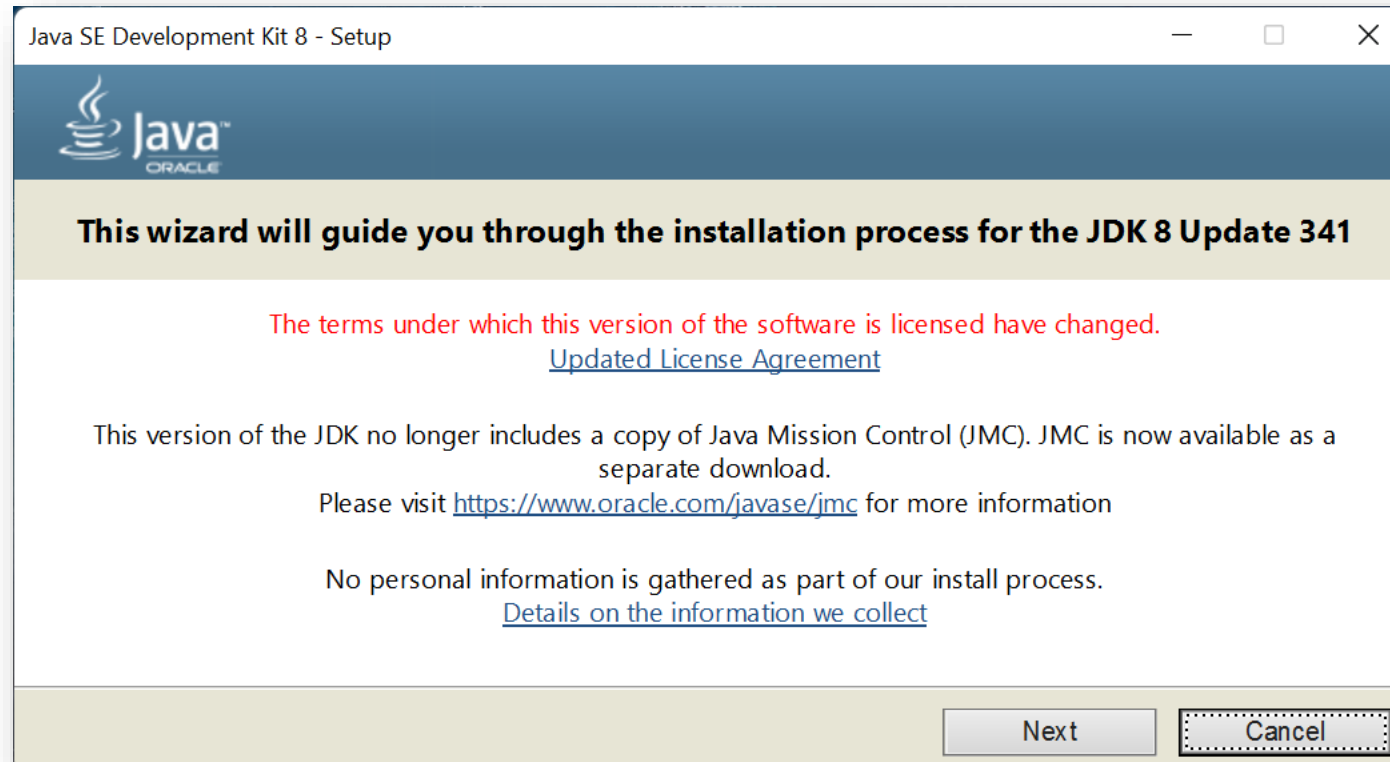
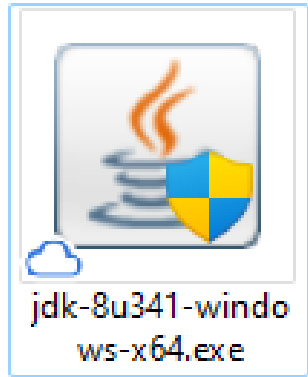
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NOTE: You need to create an Oracle account in order to download it

Java SDK Installation

Click on the provided JAVA sdk 1.8 installation file



Java SDK Installation



Java SE Development Kit 8 Update 301 (64-bit) - Custom Setup

Select optional features to install from the list below. You can change your choice of features after installation by using the Add/Remove Programs utility in the Control Panel

Development Tools

Source Code

Public JRE

Feature Description

Java SE Development Kit 8 Update 301 (64-bit) JavaFX SDK and a p This will require 180 hard drive.

Install to:
C:\Program Files\Java\jdk1.8.0_301\

Next >

Java SE Development Kit 8 Update 301 (64-bit) - Complete

Java SE Development Kit 8 Update 301 (64-bit) Successfully Installed

Click Next Steps to access tutorials, API documentation, developer guides, relea and more to help you get started with the JDK.

Next Steps

Close

C:\Program Files\Java\jdk1.8.0_341

☐ Name

bin

include

jre

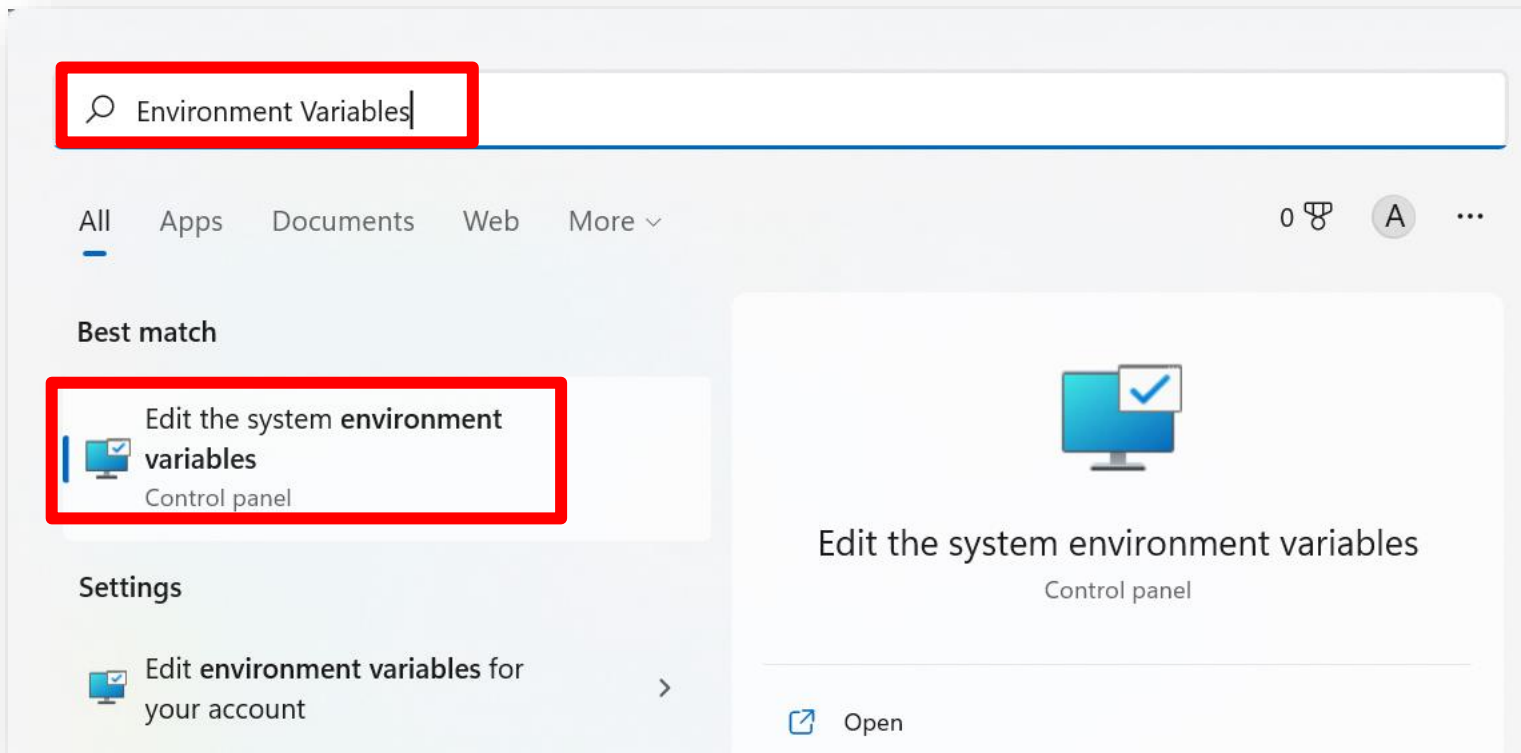
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lib

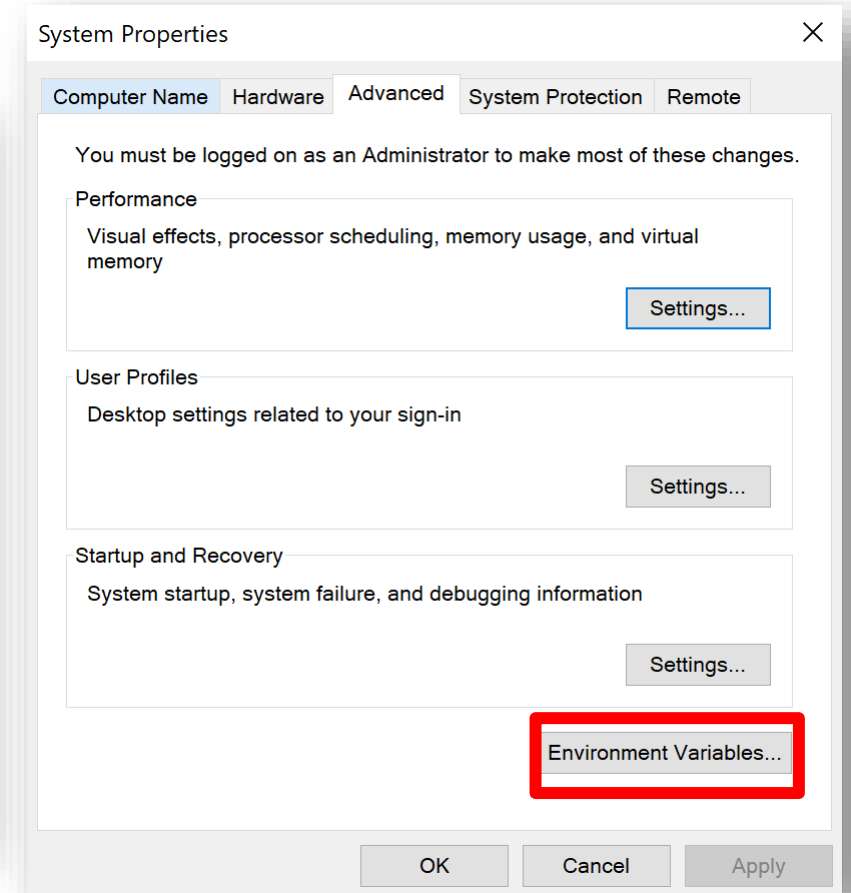
Java SDK Installation



Update System path for Java SDK



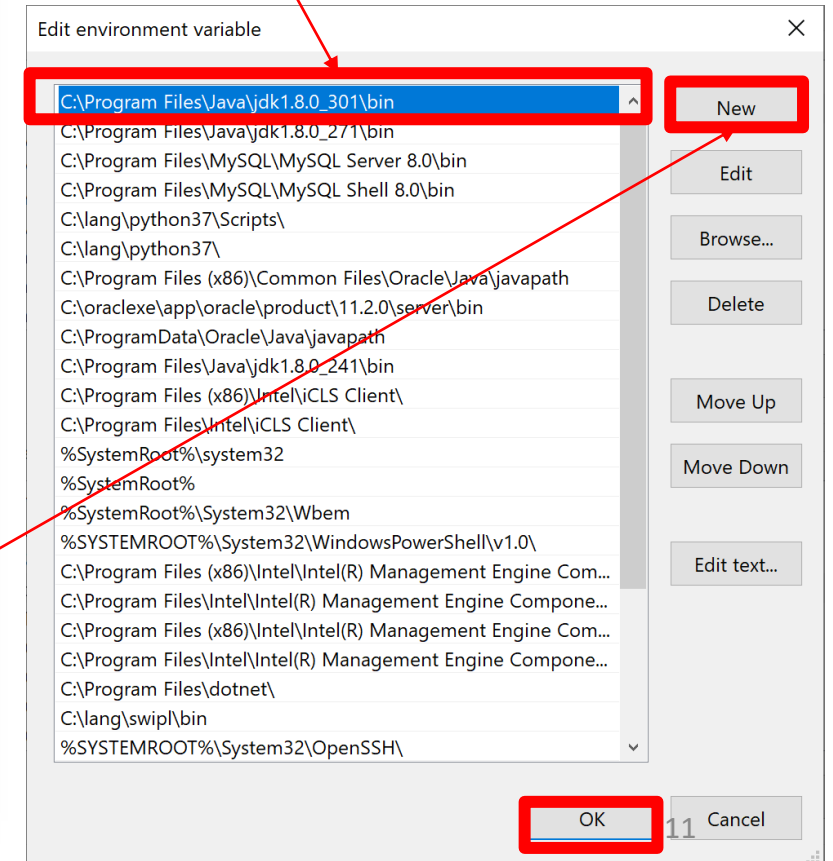
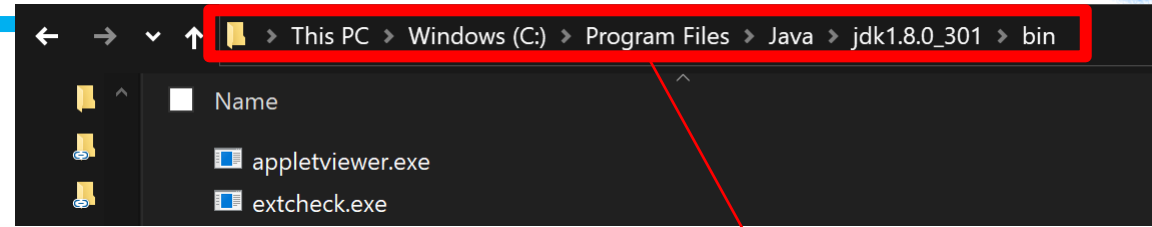
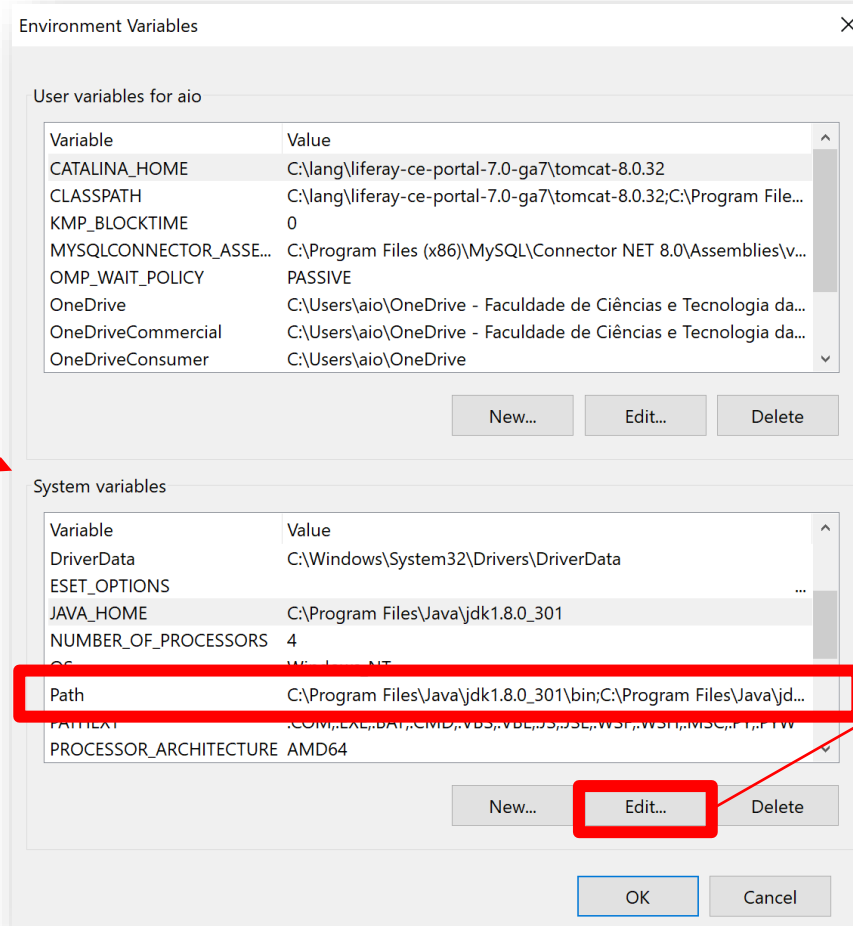
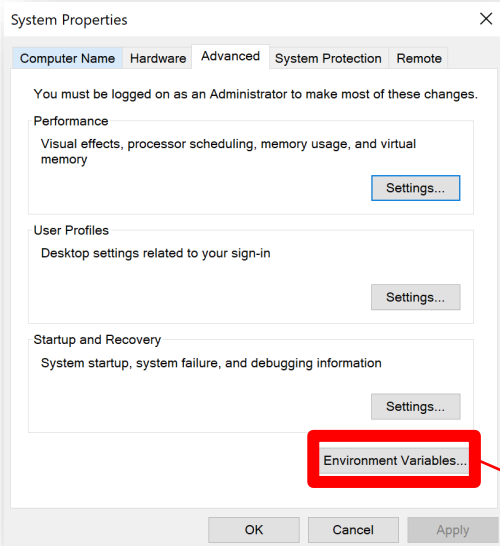
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Java SDK Installation

Click on **Environment Variables, PATH** and **Edit**

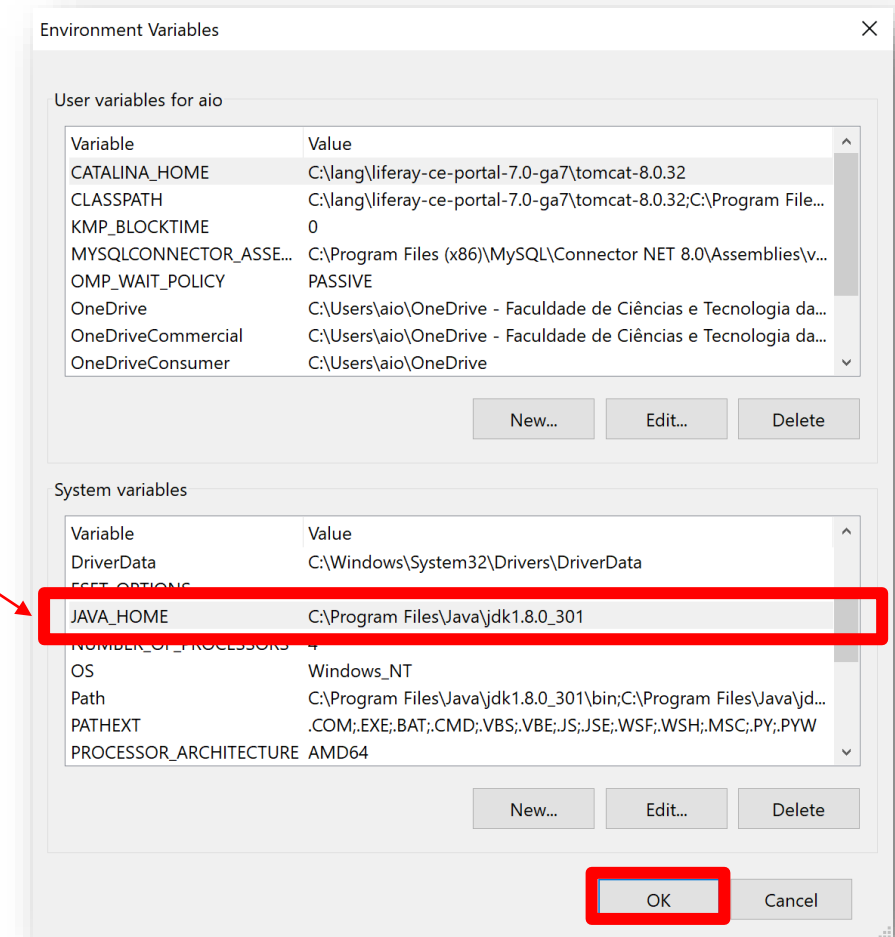
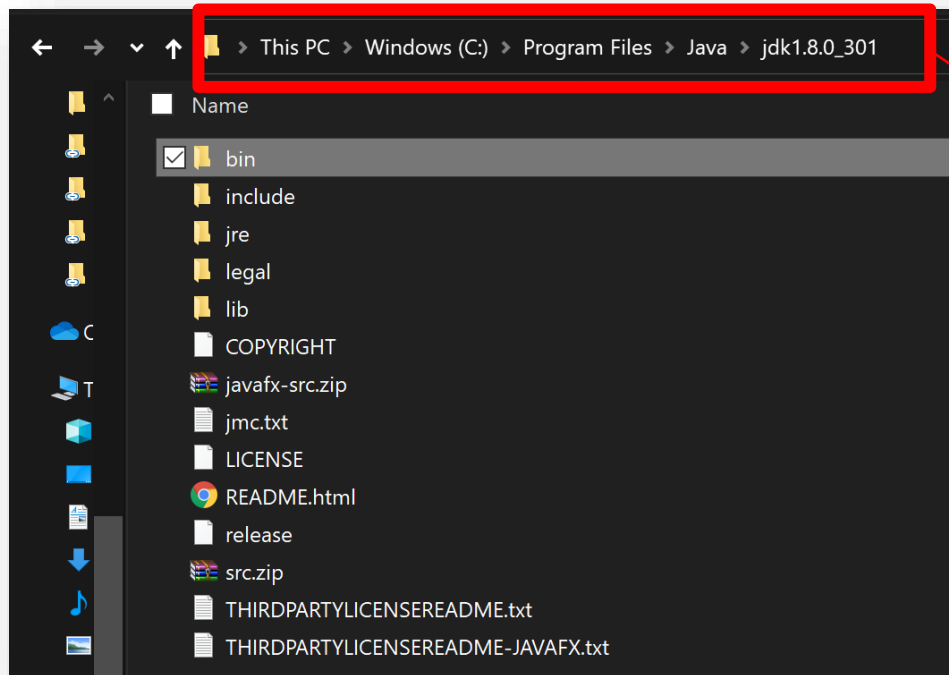


Java SDK Installation



Repeat for JAVA_HOME variable

Perform the previous steps for **JAVA_HOME** variable
(if it doesn't exist create it)
(path the same as previous, but without "bin" sub-folder):



Java SDK Installation



Testing JDK

Run a Command Prompt Window (in the search button, write down “cmd.exe”)

Inside the prompt, use the command:

and {
 java -version
 javac -version

```
Command Prompt
Microsoft Windows [Version 10.0.18363.1556]
(c) 2019 Microsoft Corporation. All rights reserved.

C:\Users\airo>java -version
java version "1.8.0_301"
Java(TM) SE Runtime Environment (build 1.8.0_301-b09)
Java HotSpot(TM) 64-Bit Server VM (build 25.301-b09, mixed mode)

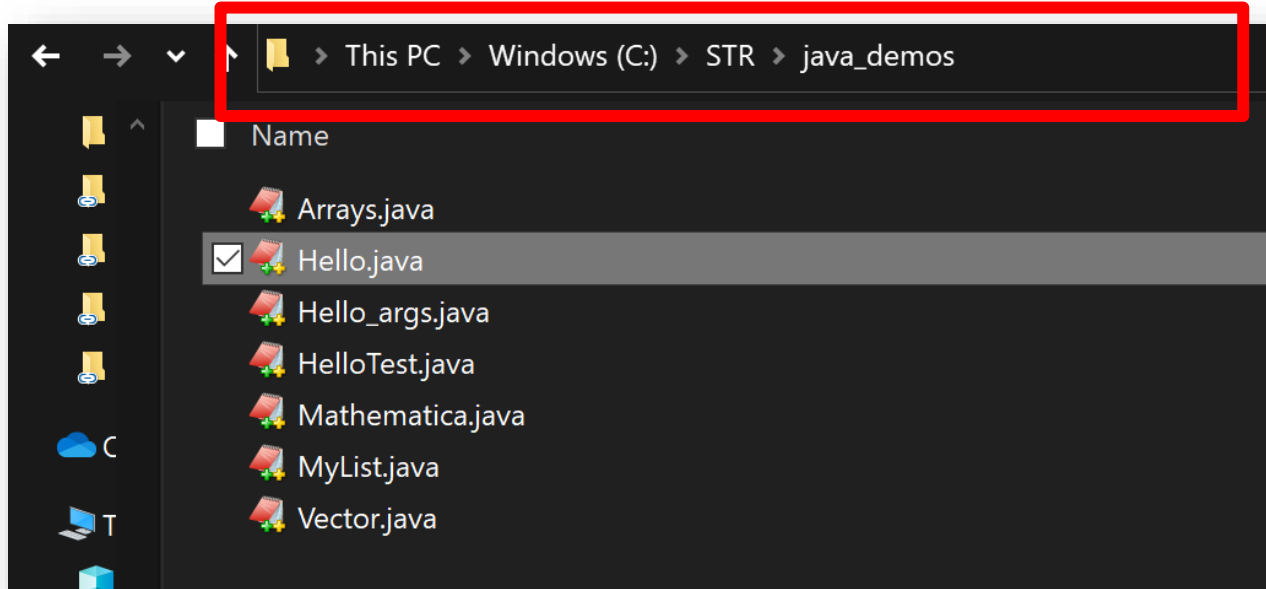
C:\Users\airo>javac -version
javac 1.8.0_301

C:\Users\airo>
```



If the result is similar to the figure, then **SUCCESS**.

Extract java_demo.zip to c:\str



Example 1 – My first java program

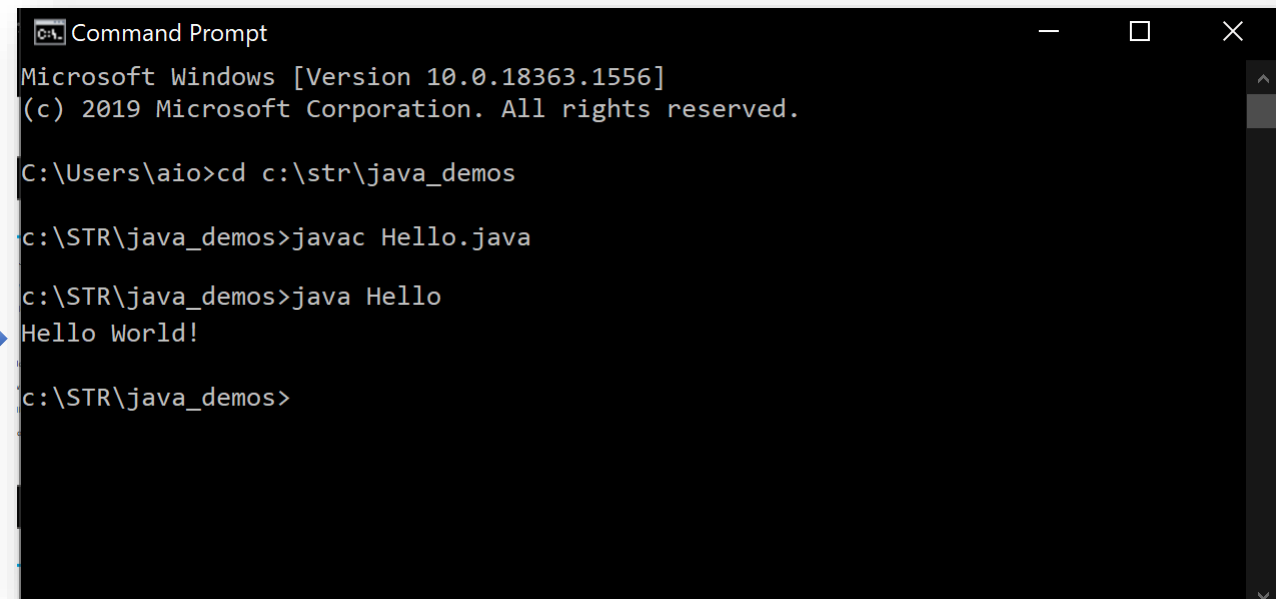
In `c:\str\java_demos` edit the file named **Hello.java**

```
1 class Hello
2 {
3     public static void main(String[] args)
4     {
5         System.out.println("Hello World!");
6     }
7 }
```

Enter a console window and go to `c:\str\java_demos`

Compile with: **javac Hello.java**

Execute with: **java Hello**



```
Command Prompt
Microsoft Windows [Version 10.0.18363.1556]
(c) 2019 Microsoft Corporation. All rights reserved.

C:\Users\ain>cd c:\str\java_demos

c:\STR\java_demos>javac Hello.java

c:\STR\java_demos>java Hello
Hello World!

c:\STR\java_demos>
```

Example 2 – Using arguments in method “main”

In `c:\str\java_demos` edit the file named `Hello_args.java`

```
1 public class Hello_args
2 {
3     public static void main(String[] args){
4         System.out.println("-----");
5         for (int i=0; i<args.length; i++){
6
7             System.out.printf("\nHello arg[%d] = %s", i ,args[i]);
8
9             System.out.printf("\nHello arg[" + i + "] = " + args[i] );
10
11     }
12     System.out.println("\n-----");
13 }
14 }
```

Compile with: `javac Hello_args.java`

Execute with: `java Hello_args`

and `java Hello_args I like JAVA 1 2 3 !`

```
Command Prompt
c:\STR\java_demos>
c:\STR\java_demos>javac Hello_args.java

c:\STR\java_demos>java -cp . Hello_args I like JAVA 1 2 3 !
-----
Hello arg[0] = I
Hello arg[0] = I
Hello arg[1] = like
Hello arg[1] = like
Hello arg[2] = JAVA
Hello arg[2] = JAVA
Hello arg[3] = 1
Hello arg[3] = 1
Hello arg[4] = 2
Hello arg[4] = 2
Hello arg[5] = 3
Hello arg[5] = 3
Hello arg[6] = !
Hello arg[6] = !
-----
c:\STR\java_demos>
```

Example 3 – Using math functions

In `c:\str\java_demos` edit the file named **Vector.java**

The code at the right implements a class vector, which is not a program, but a class for defining the mathematical concept of vector.

```
1 public class Vector {
2     private double x;
3     private double y;
4     private double z;
5
6     public Vector(double x, double y, double z) {
7         this.x = x;
8         this.y = y;
9         this.z = z;
10    }
11    public void sum(double x, double y, double z){
12        this.x+= x; // this.x = this.x + x;
13        this.y+= y;
14        this.z+= z;
15    }
16    public Vector scalarProduct(Vector v) {
17        Vector result = new Vector(0,0,0);
18        result.x = this.x * v.x;
19        result.y = this.y * v.y;
20        result.z = this.z * v.z;
21        return result;
22    }
23    public double module() {
24        double module = Math.sqrt( Math.pow(this.x,2)+Math.pow(this.y,2)+Math.pow(this.z,2) );
25        return module;
26    }
27    public String toString() {
28        String s = new String(" "+this.x+"x "+this.y+"y "+this.z+"z");
29        return s;
30    }
31 }
32
```

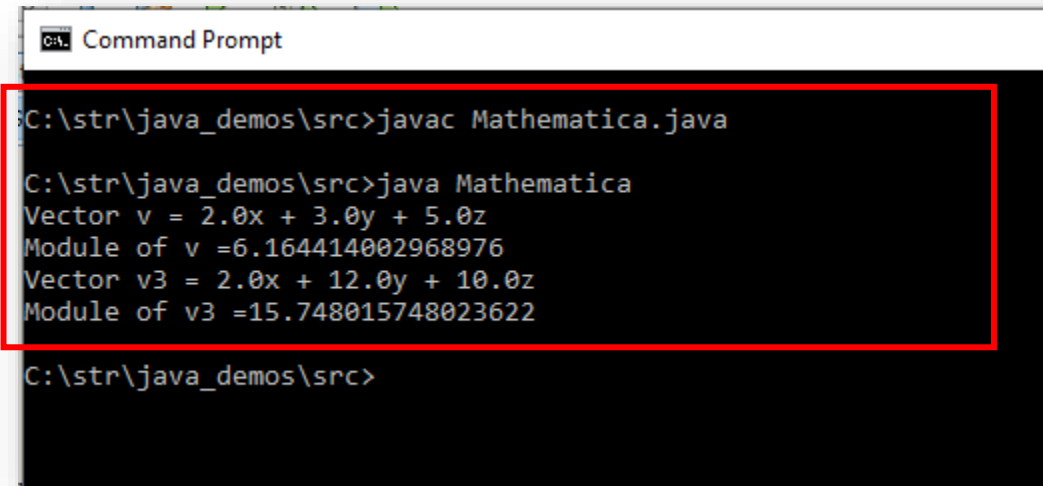
Example 3 – Using math functions

In `c:\str\java_demos` edit the file named **Mathematica.java**

```
1 public class Mathematica {  
2  
3     public Mathematica() {  
4         System.out.println("Put initialization code here");  
5     }  
6     static public void main(String args[]) {  
7  
8         Vector v = new Vector(2,3,5);  
9  
10        System.out.println("Vector v =" + v.toString());  
11        System.out.println("Module of v =" + v.module());  
12  
13        Vector v2 = new Vector(1,4,2);  
14        Vector v3 = v.scalarProduct(v2);  
15  
16        System.out.println("Vector v3 =" + v3);  
17        System.out.println("Module of v3 =" + v3.module() );  
18    }  
19 }  
20
```

It uses the code of class `Vector` for performing operations with vectors.

Modularity is one of the great OOP features existing in languages like JAVA.



```
C:\str\java_demos\src>javac Mathematica.java  
  
C:\str\java_demos\src>java Mathematica  
Vector v = 2.0x + 3.0y + 5.0z  
Module of v =6.164414002968976  
Vector v3 = 2.0x + 12.0y + 10.0z  
Module of v3 =15.748015748023622  
  
C:\str\java_demos\src>
```

Example 4 – Using arrays

In `c:\str\java_demos` edit the file named `Arrays.java`

```
Command Prompt
Microsoft Windows [Version 10.0.22000.1098]
(c) Microsoft Corporation. All rights reserved.
```

```
C:\Users\ao>cd c:\str\java_demos

c:\str\java_demos>javac Arrays.java

c:\str\java_demos>java Arrays
what , is , the , matrix, ?
you , must , see , it , yourself, ...
matrix={
  a, f, 1, b
  d, 5, 2, 8
  1, 3, 9, e
  f, a, c, a
}

c:\str\java_demos>
```

```
1 public class Arrays {
2
3     public void printArray(Object array[]){
4         for(int i=0; i<array.length; i++) {
5             if(i>0)
6                 System.out.print(" ");
7                 System.out.print(array[i]);
8         }
9         System.out.println();
10    }
11
12    public void printMatrix(Object matrix[][] ) {
13        System.out.println("matrix=");
14        for(int i=0; i<matrix.length; i++) {
15            System.out.print(" ");
16            printArray( matrix[i] );
17        }
18        System.out.println("}");
19    }
20
21    static public void main(String args[]) {
22        String array1[]={"what ", "is ", "the ", "matrix", "?"};
23        String array2[]={"you ", "must ", "see ", "it ", "yourself", "..."};
24        String matrix[][] = {
25            {"a", "f", "1", "b"},
26            {"d", "5", "2", "8"},
27            {"1", "3", "9", "e"},
28            {"f", "a", "c", "a"}
29        };
30        Arrays d = new Arrays();
31        d.printArray(array1);
32        d.printArray(array2);
33        d.printMatrix(matrix);
34    }
35 }
```

Example 5 – Using linked lists

In `c:\str\java_demos` edit the file named `MyList.java`

```
Command Prompt

c:\str\java_demos>javac MyList.java
Note: MyList.java uses unchecked or unsafe operations.
Note: Recompile with -Xlint:unchecked for details.

c:\str\java_demos>java MyList
-----
first(list)=aaa
last (list)=zzz
-----
list[0]=aaa
list[1]=ccc
list[2]=222
list[3]=100
list[4]=zzz
-----

c:\str\java_demos>
```

```
1 public class MyList {
2
3     public MyList() {
4         System.out.println("Object of MyList just created...");
5     }
6     static public void main(String args[]) {
7         java.util.LinkedList list = new java.util.LinkedList();
8
9         list.add("aaa");
10        list.add("ccc");
11        list.add("222");
12        list.add( new Integer(100) );
13        list.add("zzz");
14
15        System.out.println("-----");
16        System.out.println( "first(list)=" +list.getFirst());
17        System.out.println( "last (list)=" +list.getLast());
18        System.out.println("-----");
19
20        for(int i=0; i<list.size(); i++) {
21            System.out.println( "list["+i+"]="+list.get(i) );
22        }
23        System.out.println("-----");
24    }
25 }
```

More Info...

Go to next theoretical class for a
complete real-time java tutorial

31.10.2022



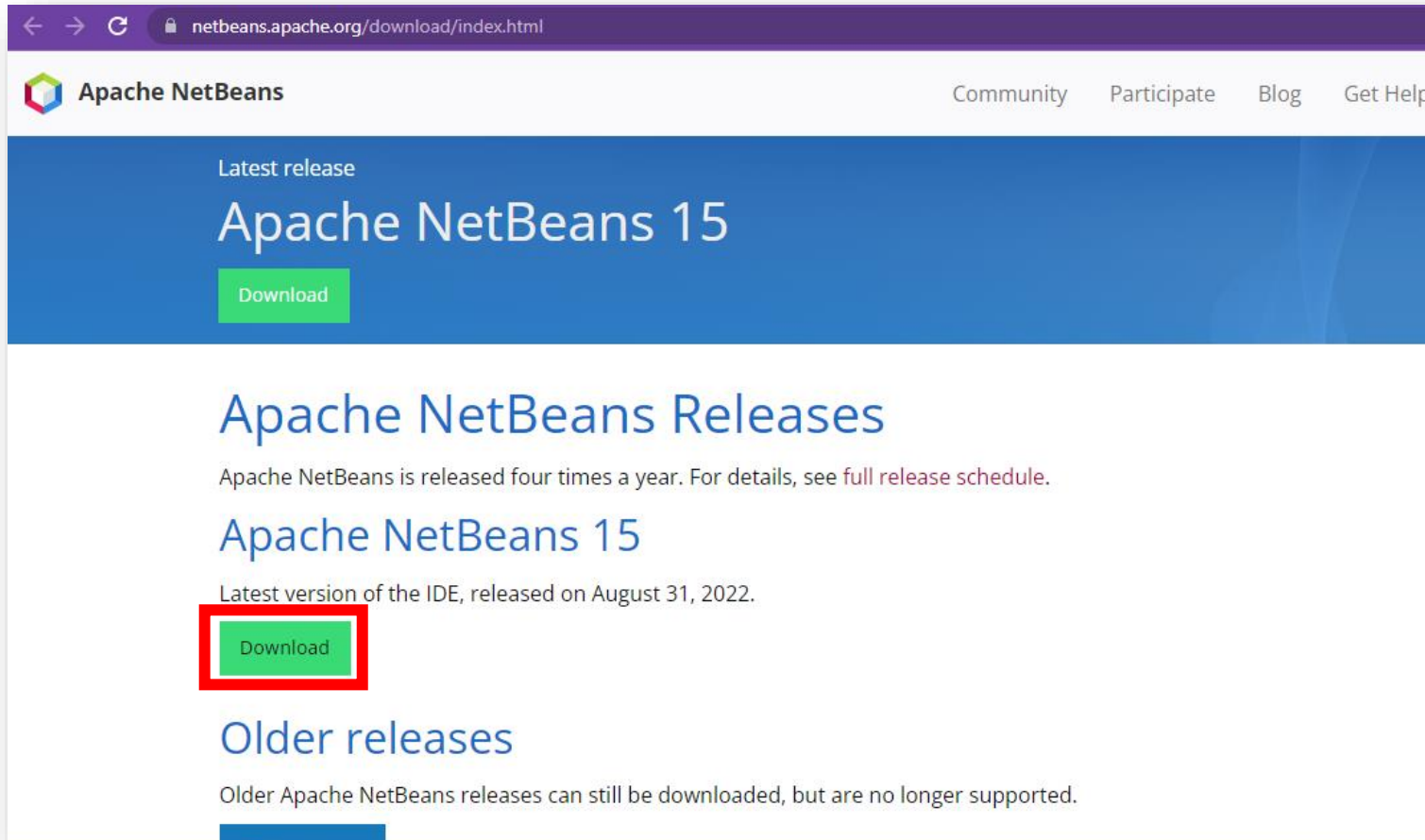


NetBeans

Netbeans Installation



Download Netbeans fom: <https://netbeans.apache.org/download/index.html>



If you already have a previous version installed (e.g. Netbeans 8.2 or Netbeans 12.5) you can skip the next slides and go to [create project](#)



Downloading Apache NetBeans 15

Apache NetBeans 15 was released on August 31, 2022. [Go here on GitHub](#) for a list of fixed issues for Apache NetBeans 15.

Apache NetBeans 15 is available for download from your closest Apache mirror.

- Binaries: [netbeans-15-bin.zip](#) (SHA-512, PGP ASC)
- Installers:
 - [Apache-NetBeans-15-bin-windows-x64.exe](#) (SHA-512, PGP ASC)
 - [Apache-NetBeans-15-bin-macosx.dmg](#) (SHA-512, PGP ASC)
- Source: [netbeans-15-source.zip](#) (SHA-512, PGP ASC)

Officially, it is important that you [verify the integrity](#) of the downloaded files using the PGP signatures (.asc file) or a hash (.sha512 files). The PGP keys used to sign this release are available [here](#).

Apache NetBeans can also be installed as a self-contained [snap package](#) on Linux.

- Tip: Linux users: use the binary zip, [Snap](#), or our [community installers](#) (there is a .deb without JDK too). This has been discussed

[Community Installers](#)

[Deployment Platforms](#)

[Building from Source](#)

[Community Approval](#)

[Earlier Releases](#)

Choose suitable installer

Netbeans Installation



The screenshot shows the Apache NetBeans download page. At the top, there is a navigation bar with links: News, About, Make a Donation, The Apache Way, Join Us, and Downloads. Below the navigation bar is the Apache Software Foundation logo and the text "COMMUNITY-LED DEVELOPMENT 'THE APACHE WAY'". There are also links for Projects, People, Community, License, and Sponsors. A red box highlights the recommended download link: <https://dlcdn.apache.org/netbeans/netbeans-installers/15/Apache-NetBeans-15-bin-windows-x64.exe>. Below this, there are sections for "HTTP", "BACKUP SITE", and "VERIFY THE INTEGRITY OF THE FILES". A red arrow points from the highlighted link to the download file icon and name at the bottom right of the slide.

News About Make a Donation The Apache Way Join Us Downloads

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Projects People Community License Sponsors

We suggest the following site for your download:

<https://dlcdn.apache.org/netbeans/netbeans-installers/15/Apache-NetBeans-15-bin-windows-x64.exe>

Alternate download locations are suggested below.

It is essential that you verify the integrity of the downloaded file using the PGP signature (.asc file) or a hash (.md5 or .sha* file).

HTTP

<https://dlcdn.apache.org/netbeans/netbeans-installers/15/Apache-NetBeans-15-bin-windows-x64.exe>

BACKUP SITE

<https://dlcdn.apache.org/netbeans/netbeans-installers/15/Apache-NetBeans-15-bin-windows-x64.exe>

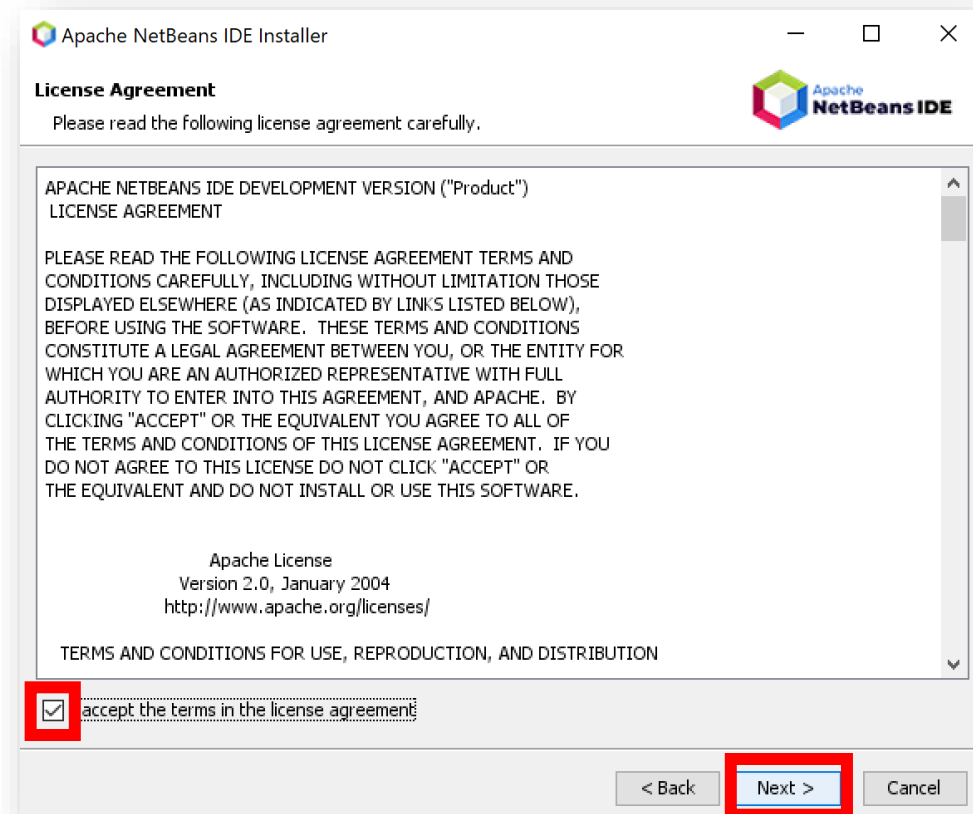
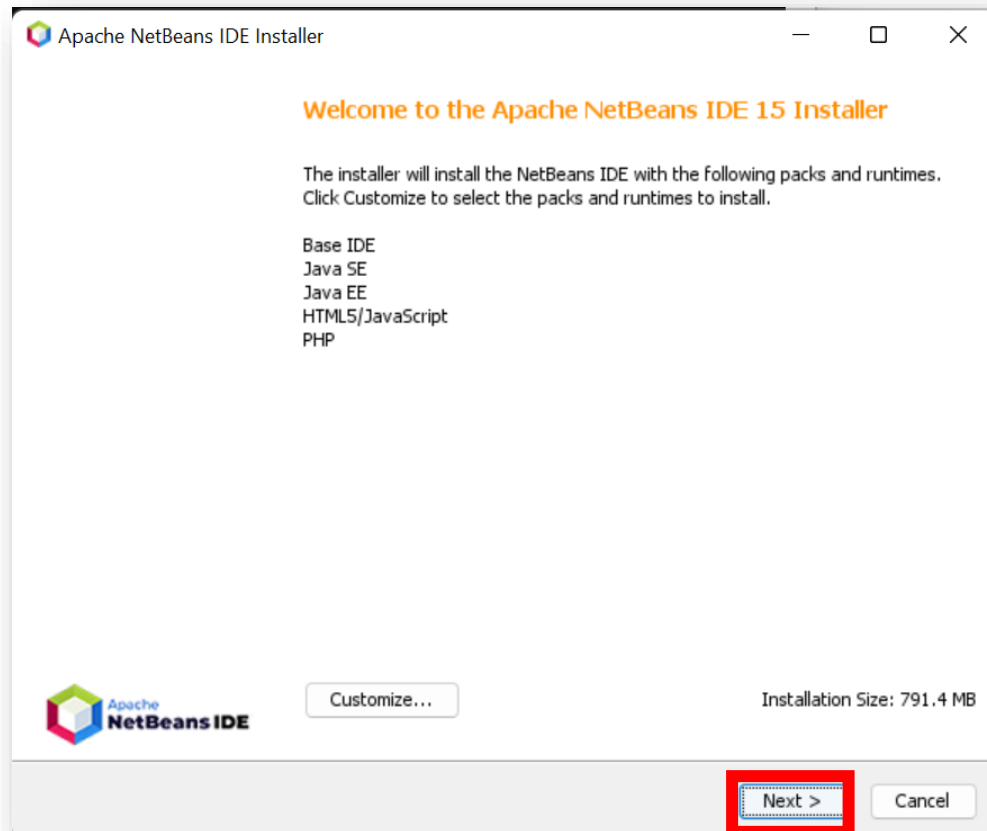
VERIFY THE INTEGRITY OF THE FILES

It is essential that you verify the integrity of the downloaded file using the PGP signature (.asc file) or a hash (.md5 or .sha* file). Please read Verifying Apache Software Foundation Releases for more information on why you should verify our releases.

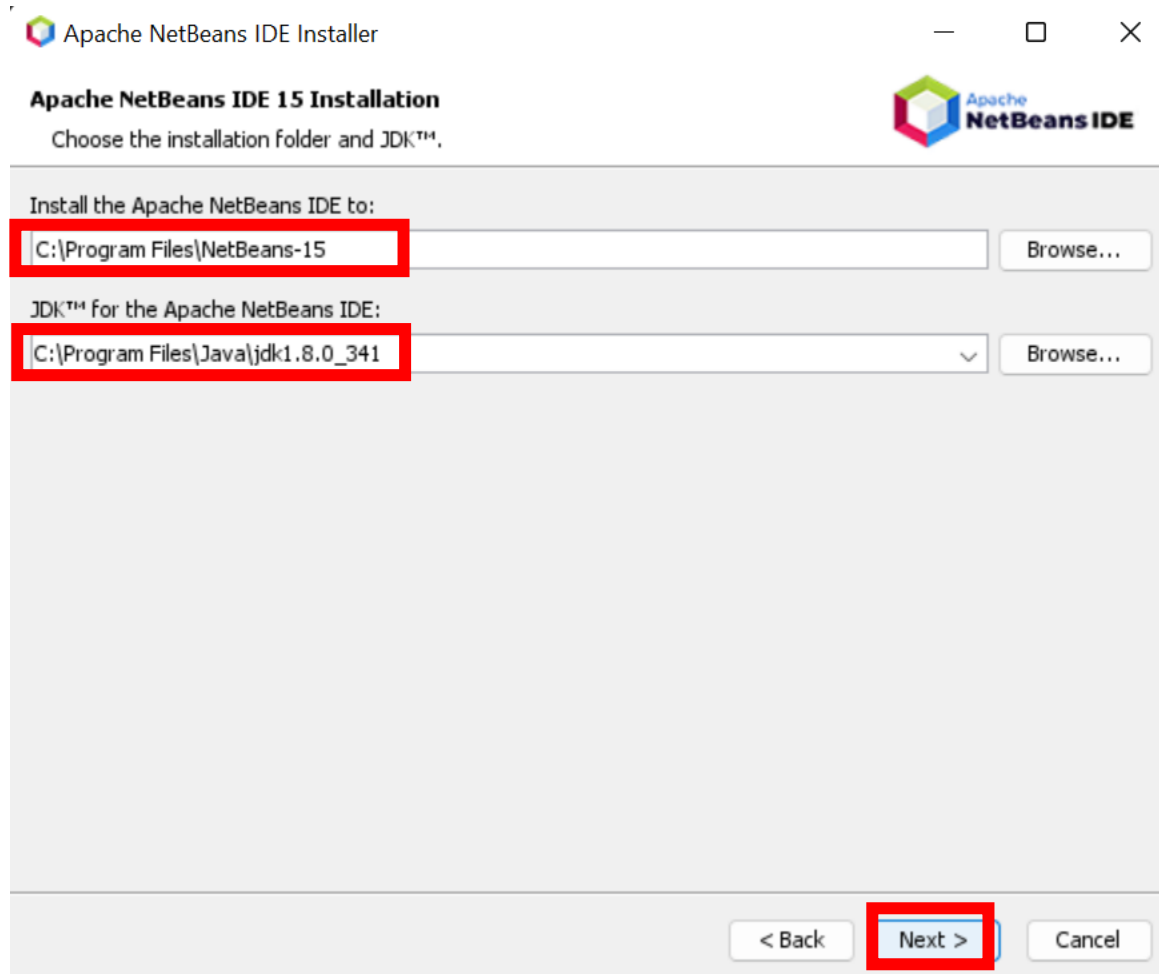
Verify the PGP signature using PGP or GPG. First download the KEYS as well as the asc signature file for the relevant distribution.

 [Apache-NetBeans-15-bin-windows-x64.exe](https://dlcdn.apache.org/netbeans/netbeans-installers/15/Apache-NetBeans-15-bin-windows-x64.exe)

Netbeans Installation

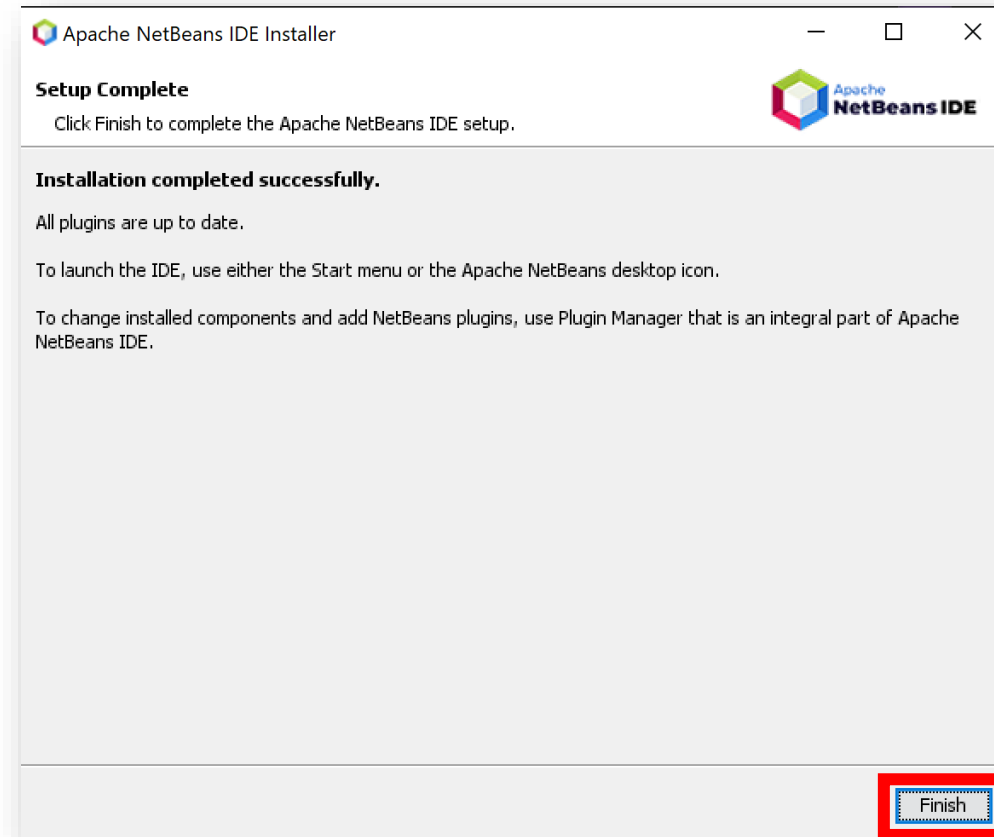
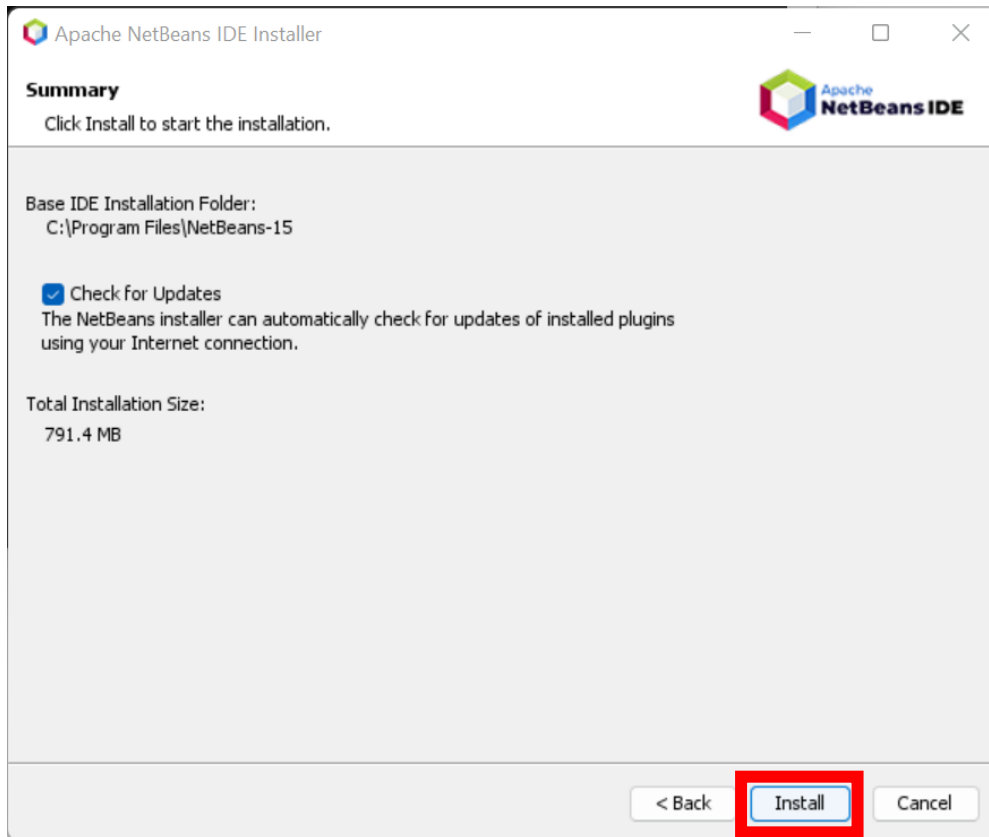


Set the install location



→ The installer should discover and set this field automatically

Netbeans Installation

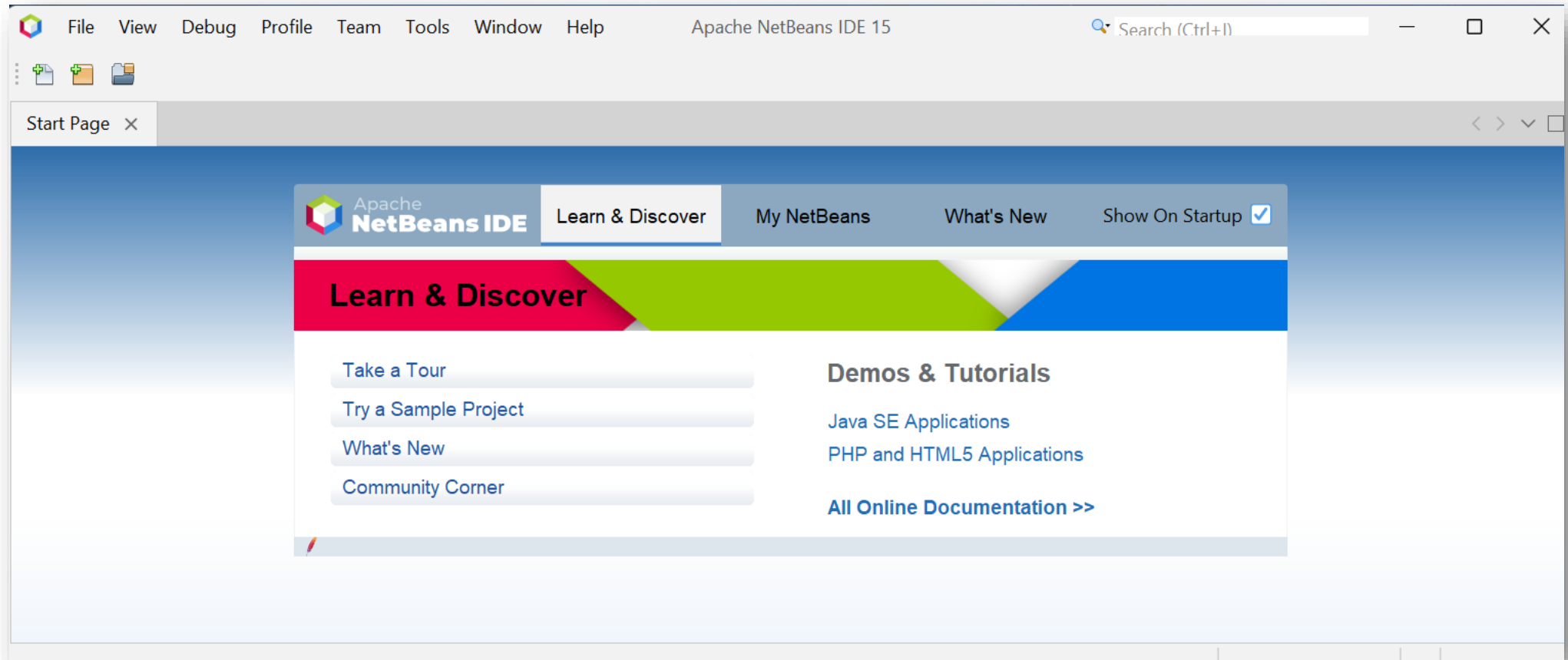


More info online (e.g. <https://netbeans.apache.org/kb/docs/java/quickstart.html>)

Netbeans Installation



Run Netbeans from “start menu”

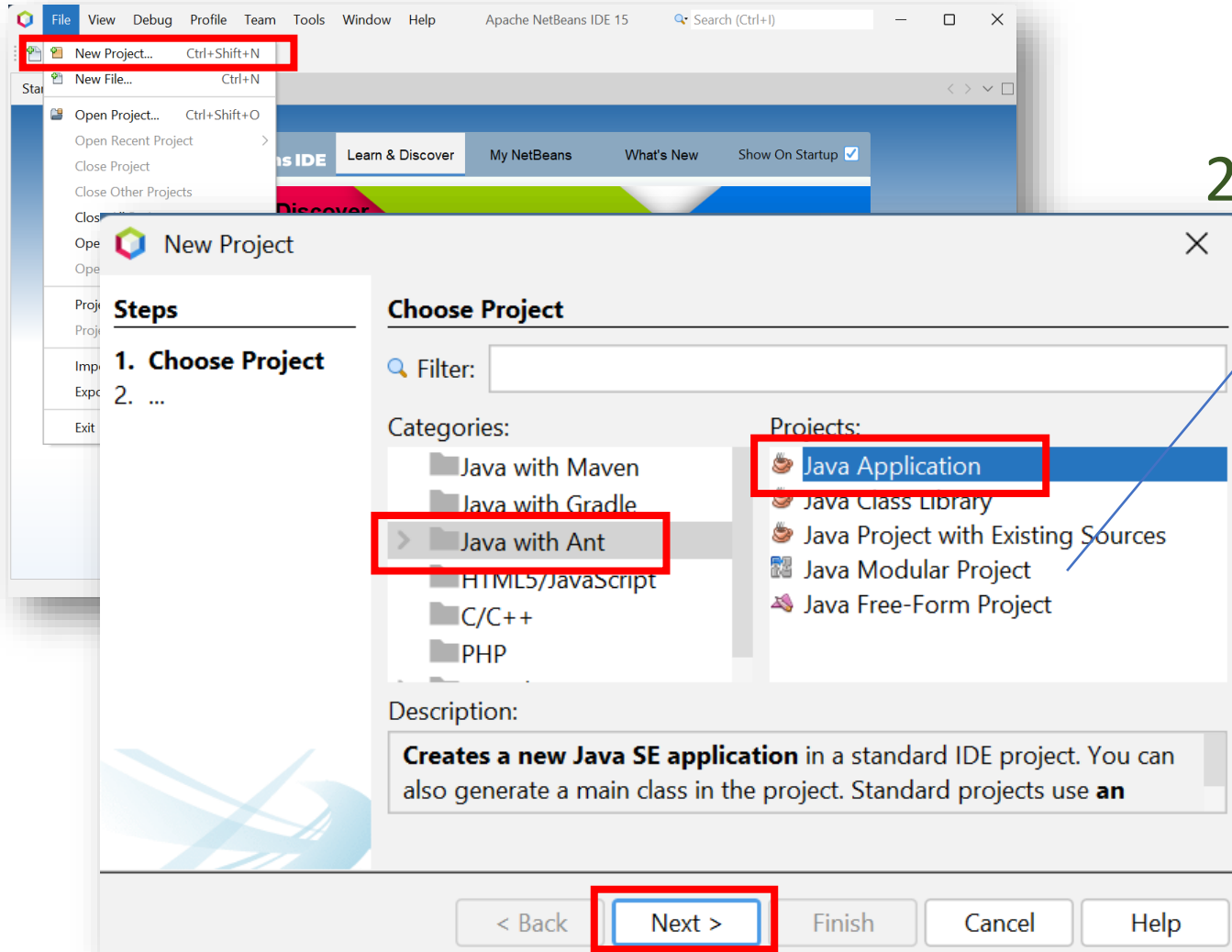


Netbeans Installation



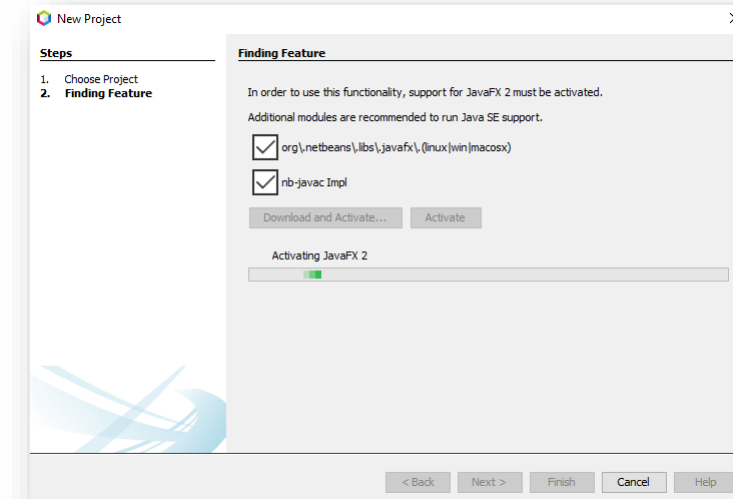
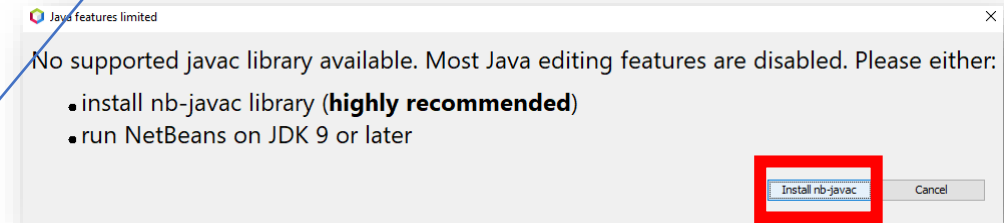
Create a new Project

1



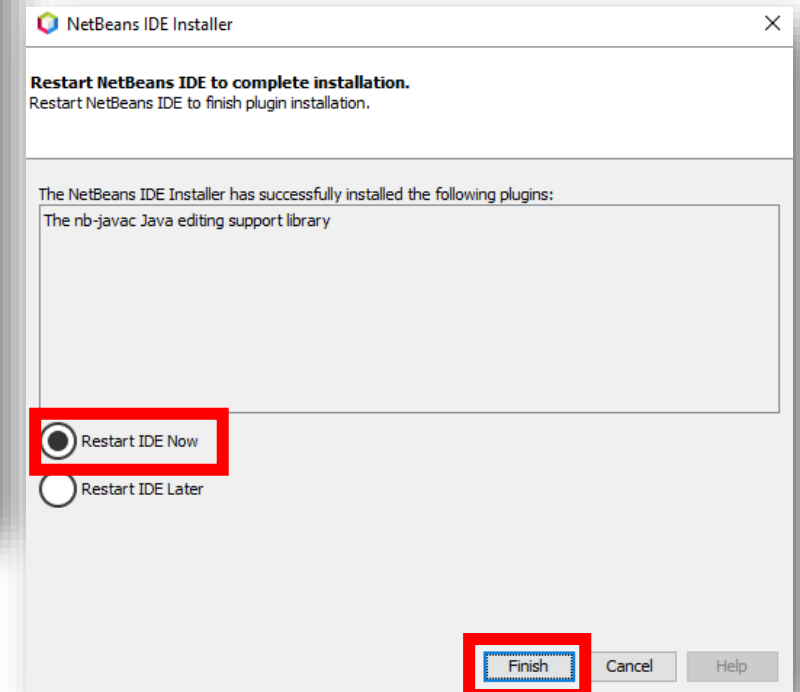
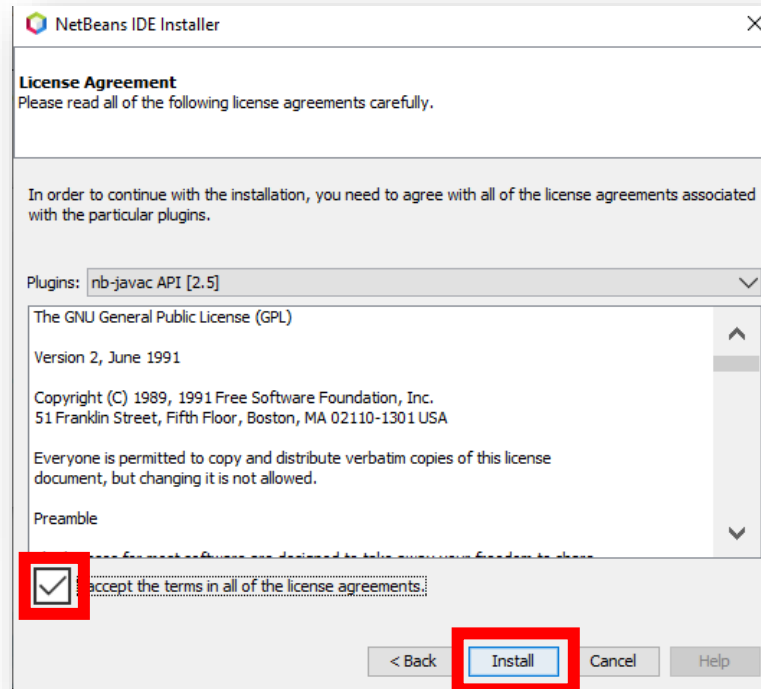
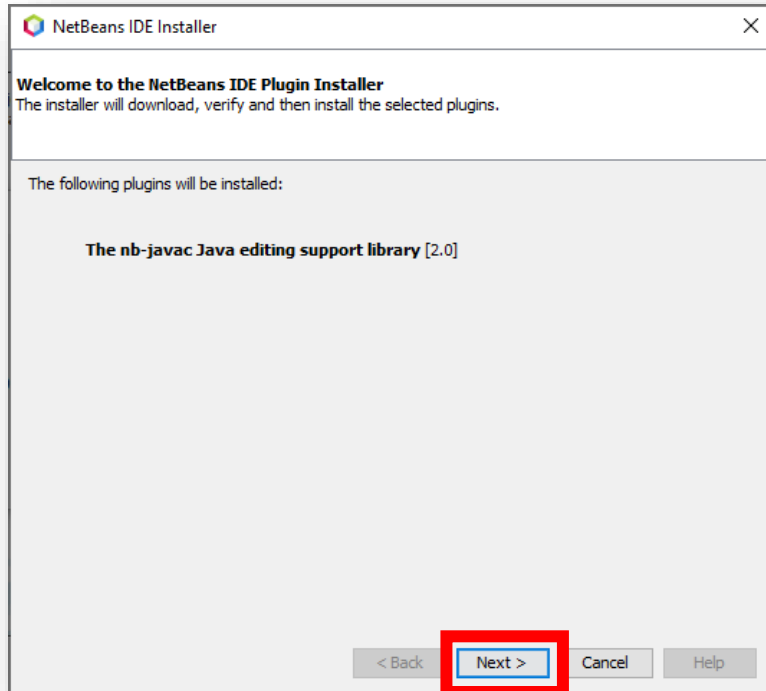
2

If some javac libraries are not available, proceed with installation....



... else go to [create a new project](#)

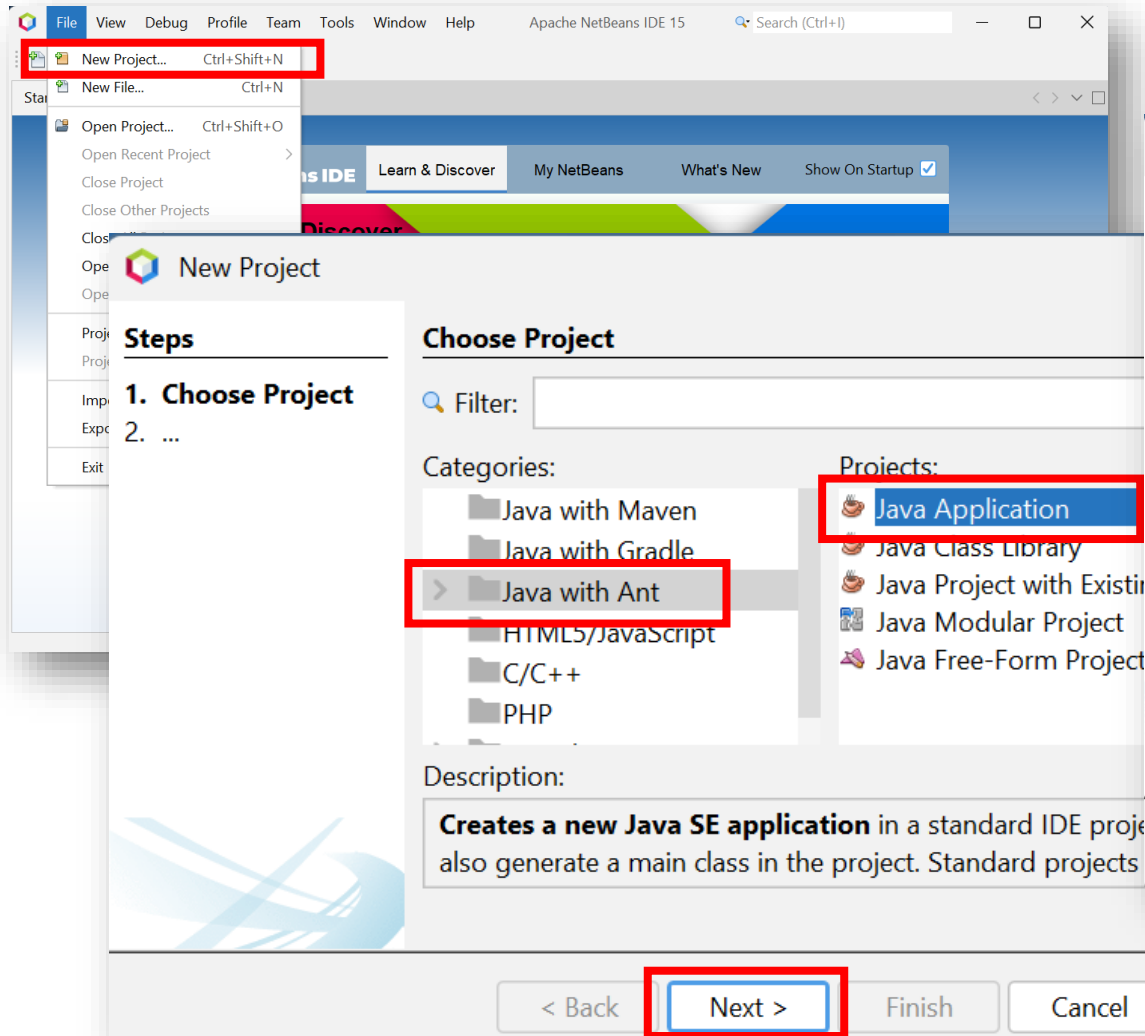
Netbeans Installation



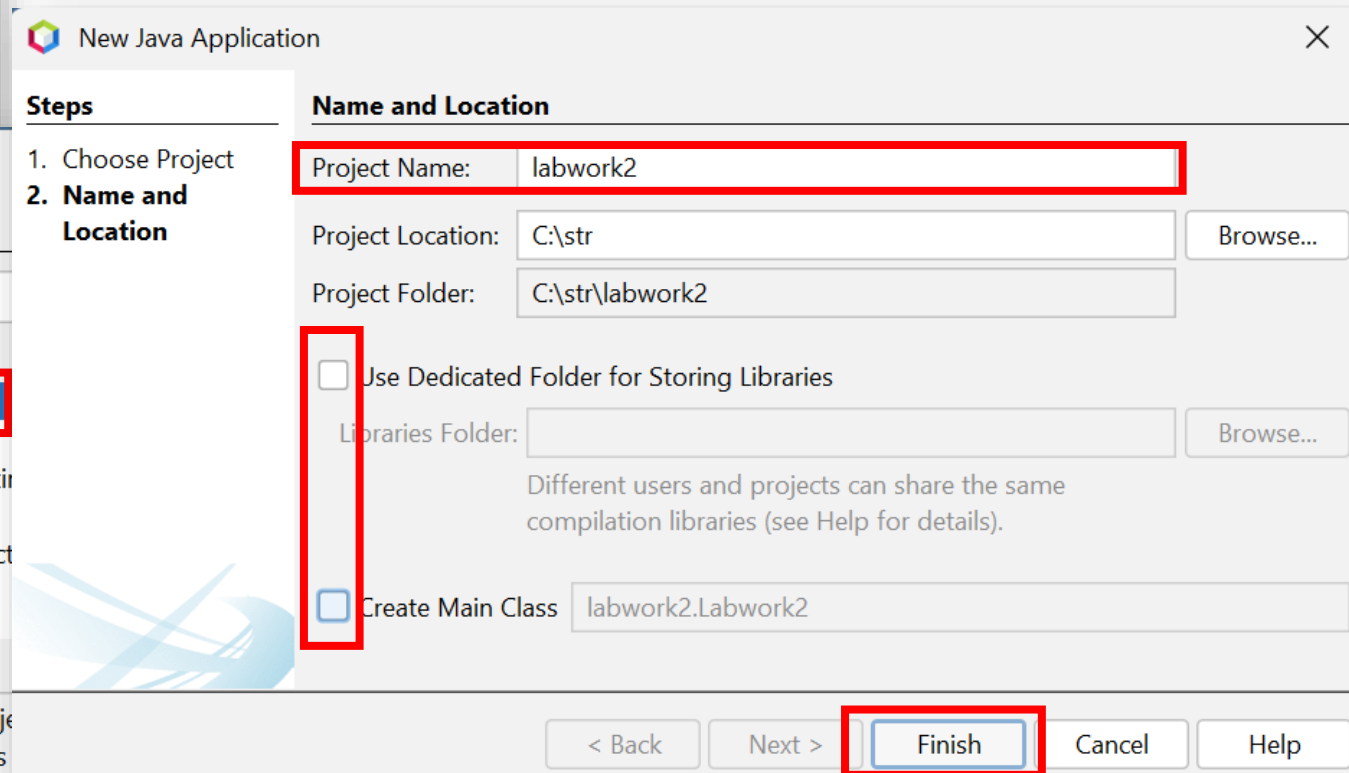
Netbeans: Create Project

Create a new Project

1



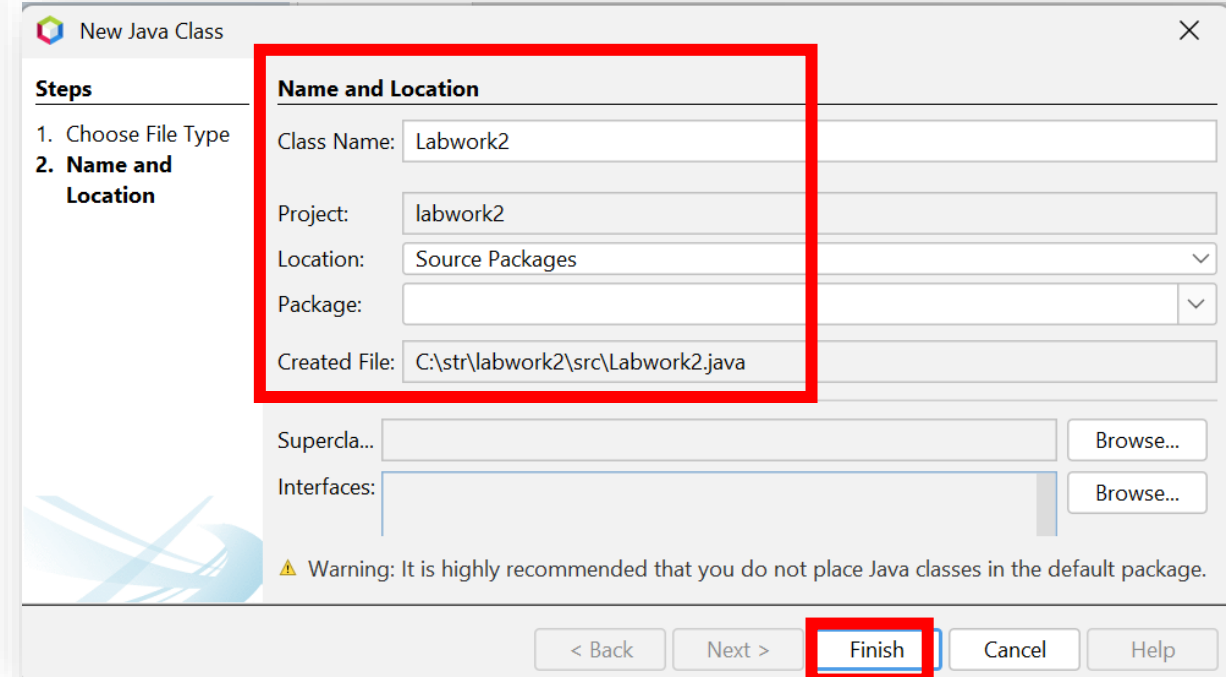
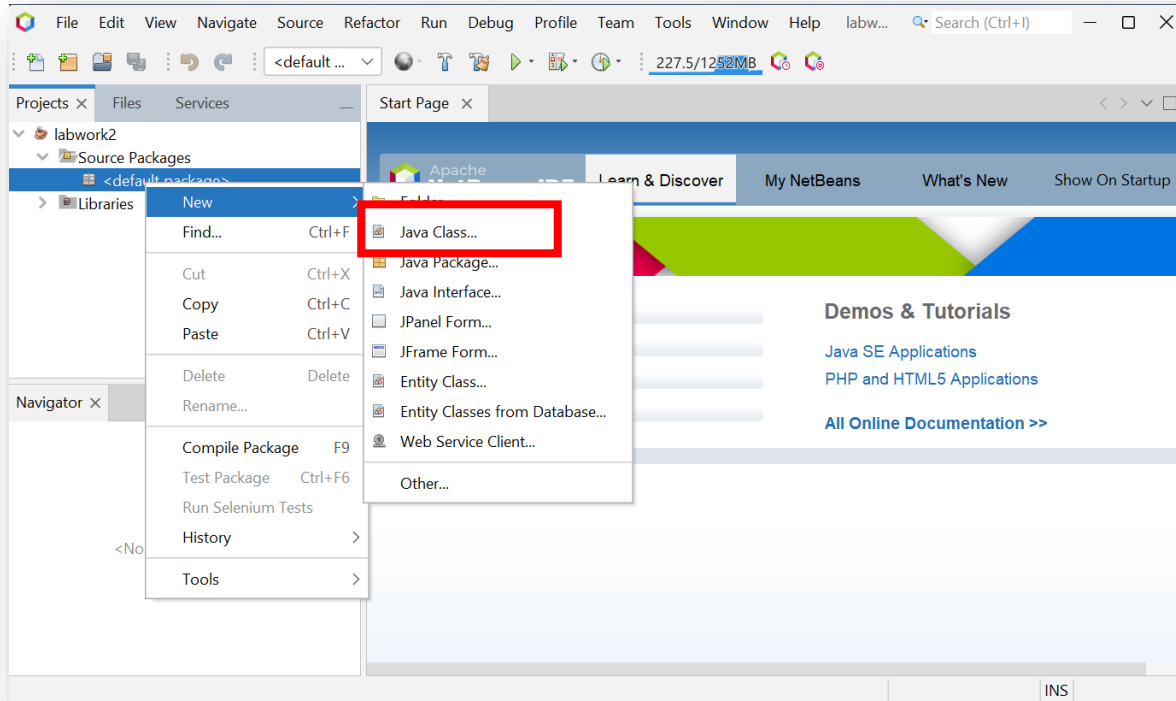
3



Netbeans: Create Project



Create class Labwork2



Create new class **Labwork2** in the **<default package>** placeholder (leave the field blank).

For now, create all classes inside the **<default package>**.

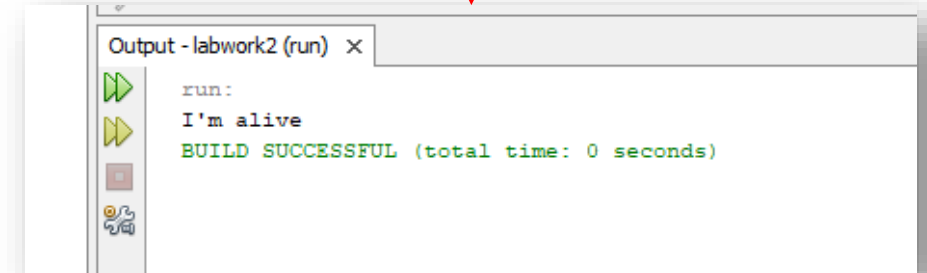
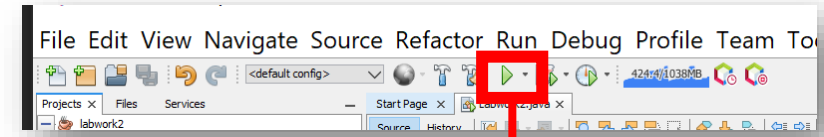
If you create them in another package, you will have to deal with a lot of technical issues later on.



Netbeans: Create Project

Create class Labwork2

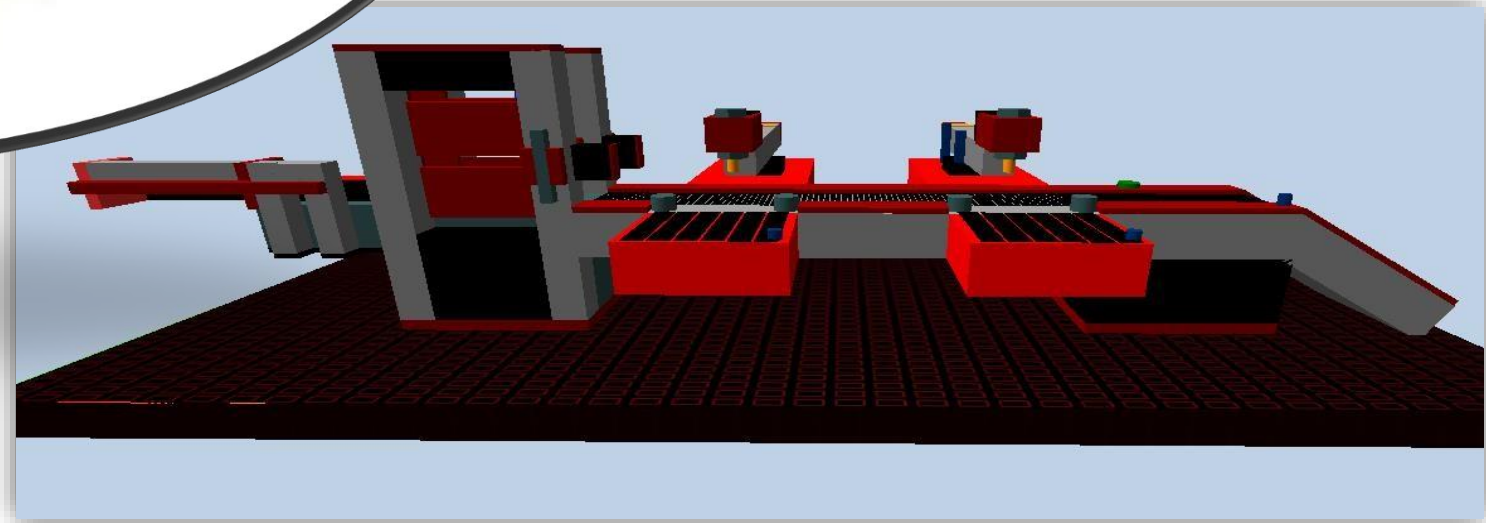
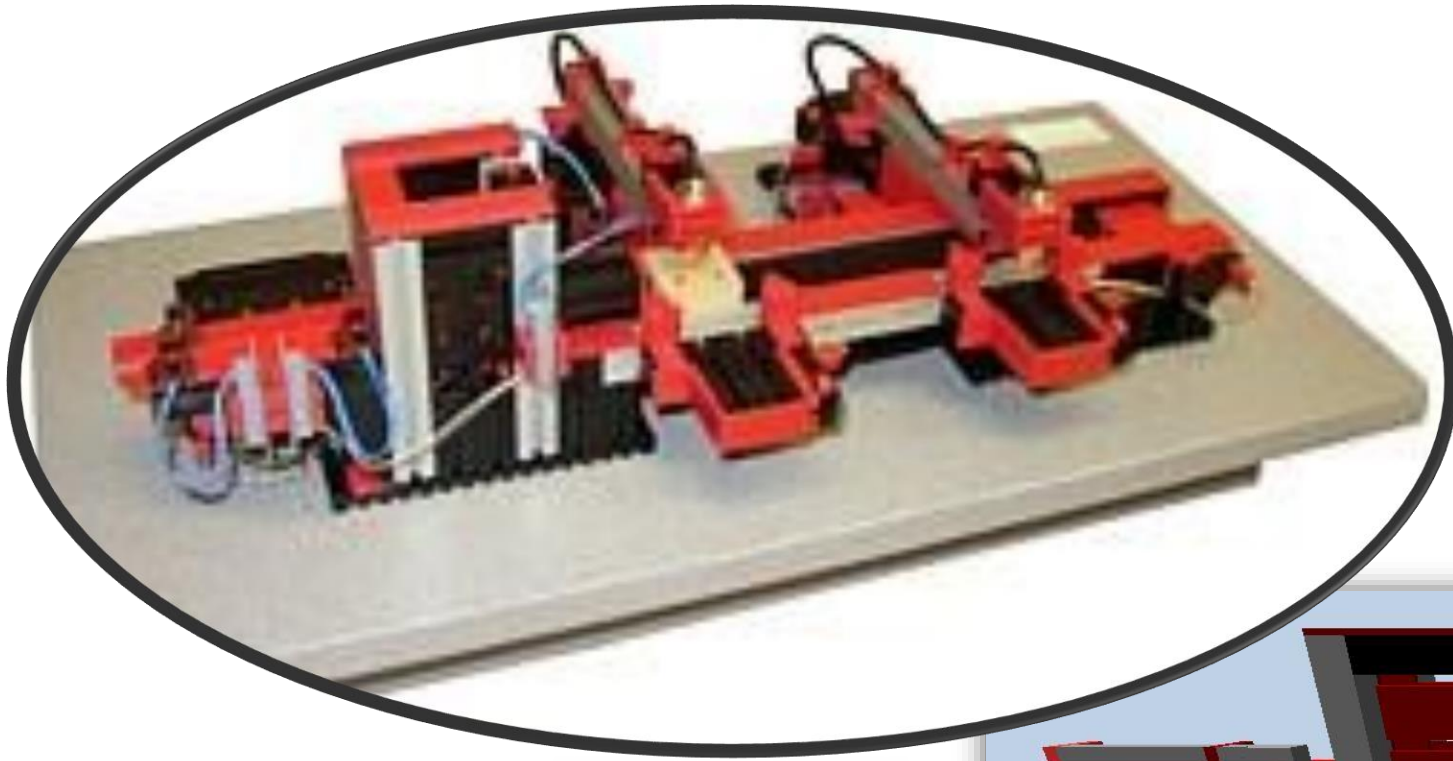
```
public class Labwork2 {  
    public static void main(String[] args) {  
        System.out.println("I'm alive");  
    }  
}
```



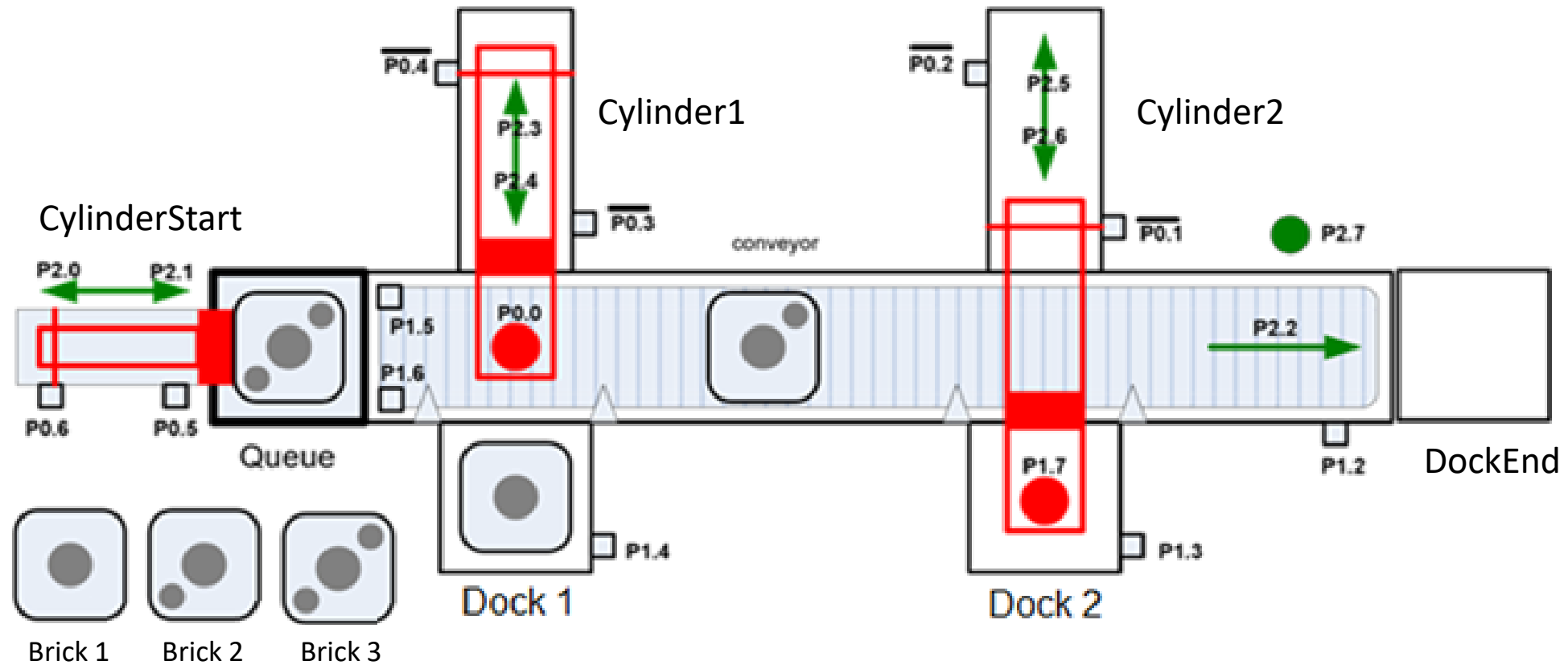
PART 2

Interaction with physical system

Introducing the kit “Splitter Conveyor”



Introducing the kit “Splitter Conveyor”

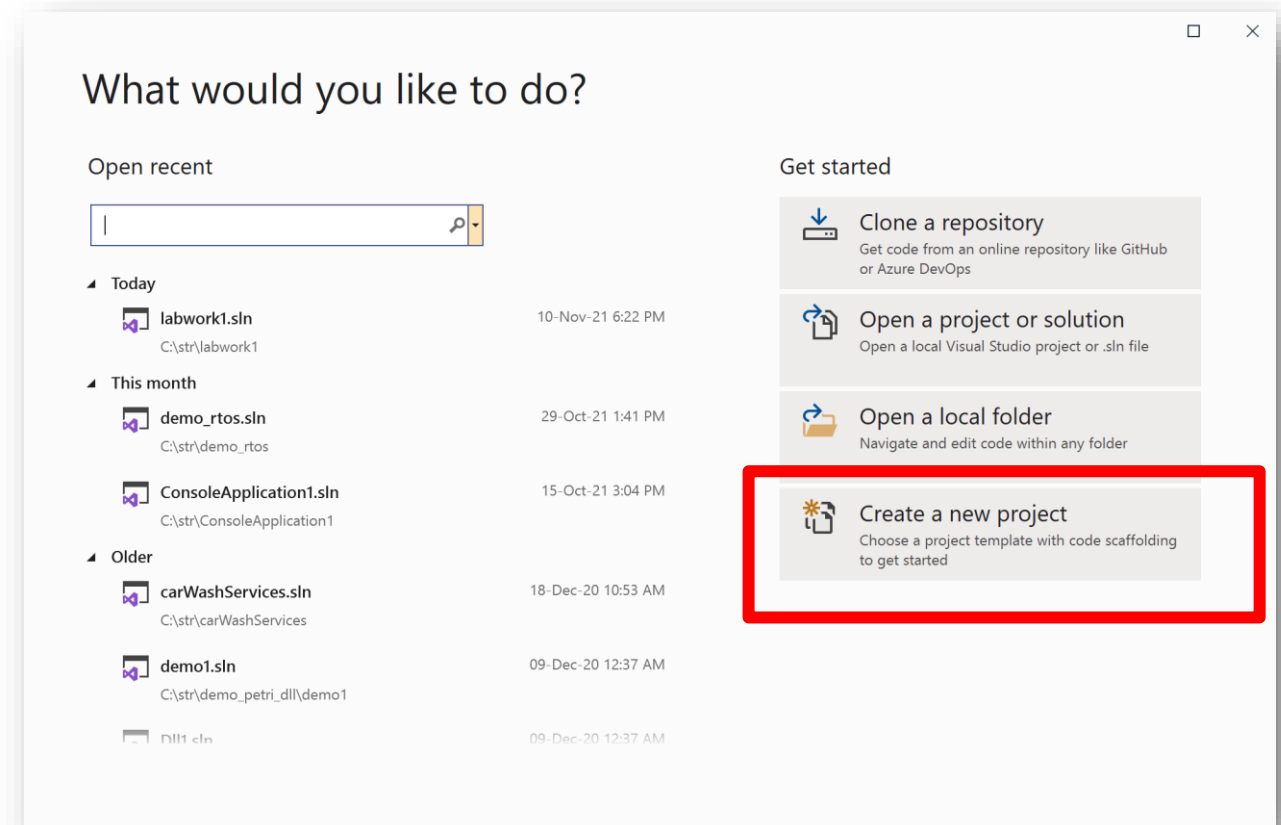


Creation of Visual studio console application project



❑ Find visual studio in your startup menu -> Run it

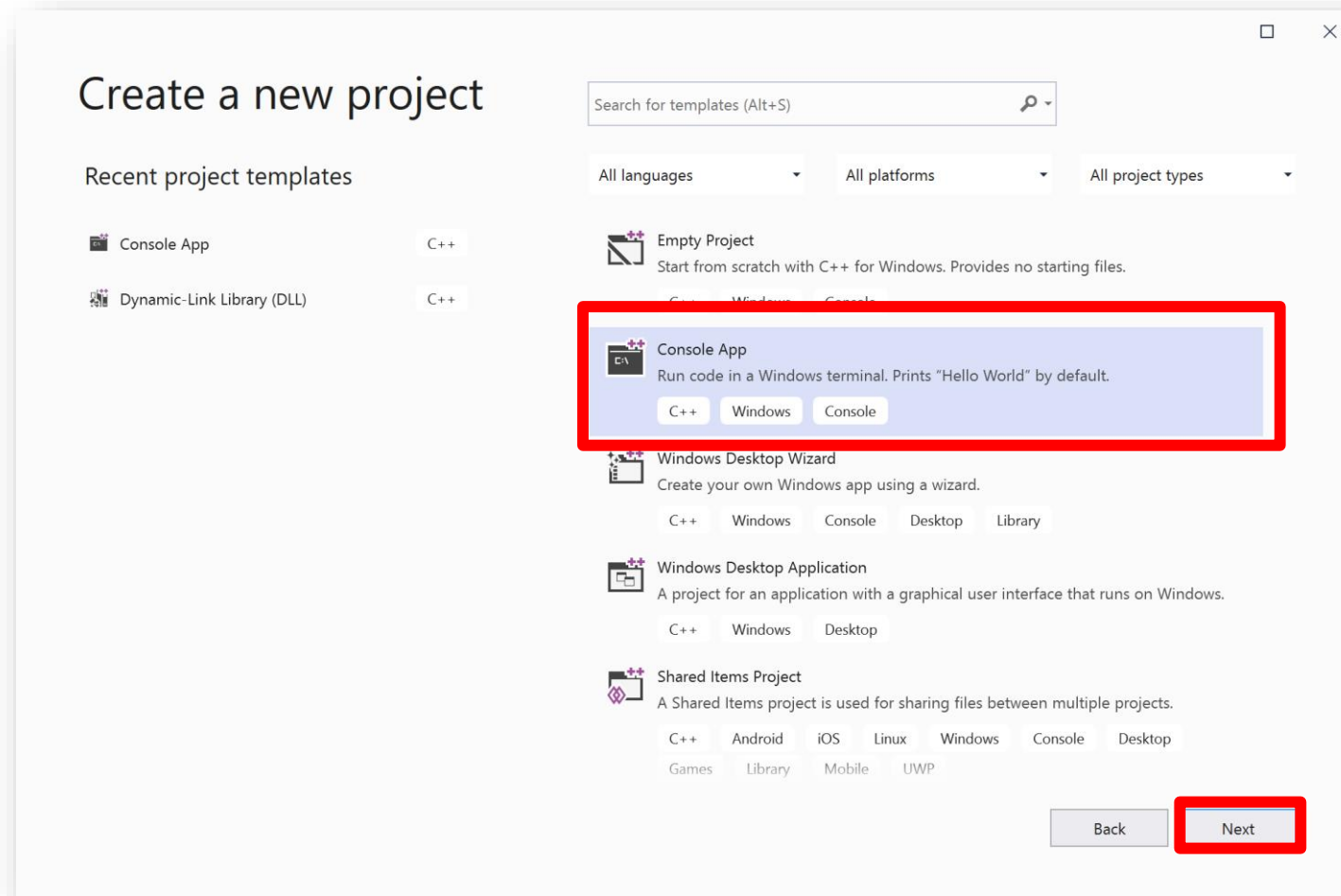
❑ Next steps are for project creation and configuration



Creation of Visual studio console application project



❑ Select Console Application



Creation of Visual studio console application project



☐ Project name: **SplitterConveyor**

☐ Location: **c:\str**

☐ Click on the checkbox

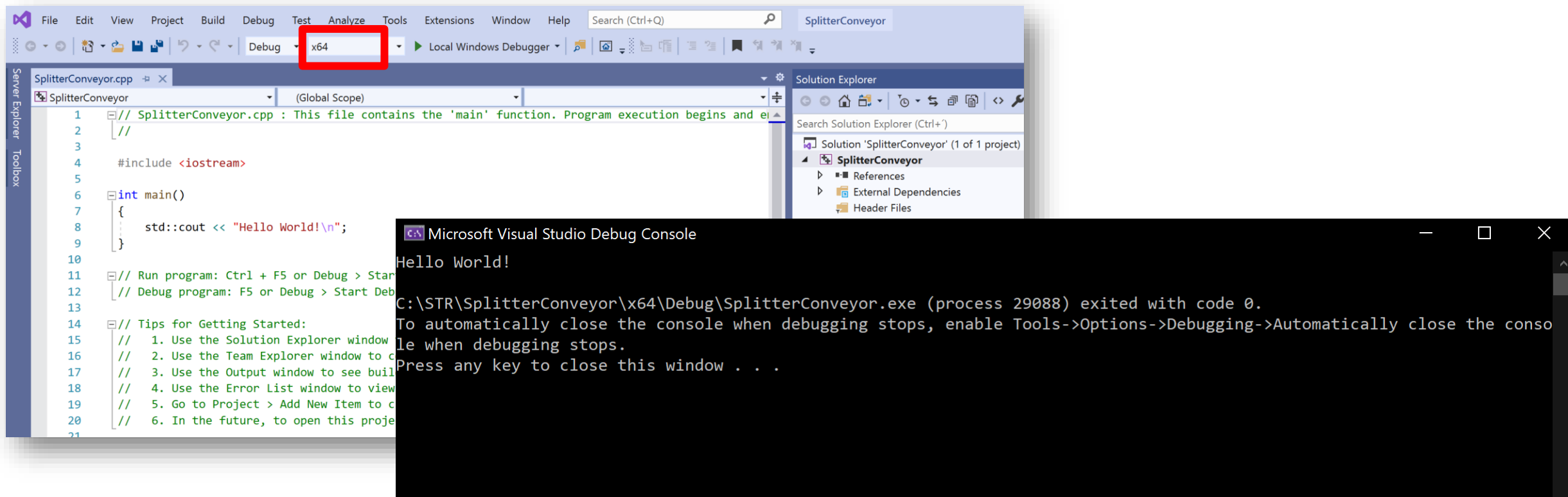
A screenshot of the Visual Studio 'Configure your new project' dialog box. The dialog has a title bar with a close button. The main content area is titled 'Configure your new project'. Below the title, there are three tabs: 'Console App', 'C++', 'Windows', and 'Console'. The 'Console App' tab is selected. Below the tabs, there are three input fields: 'Project name' with the text 'SplitterConveyor', 'Location' with the text 'C:\STR\' and a blue button with three dots to its right, and 'Solution name' with the text 'SplitterConveyor'. Below these fields, there is a checkbox labeled 'Place solution and project in the same directory' which is checked. At the bottom right of the dialog, there are two buttons: 'Back' and 'Create'. The 'Create' button is highlighted with a red rectangular border.

Creation of Visual studio console application project



Verification:

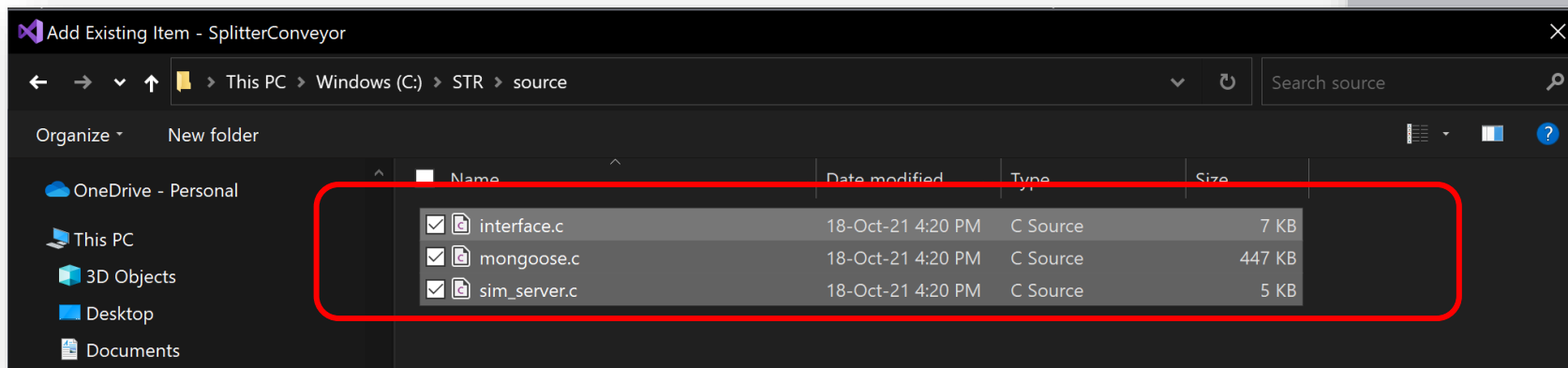
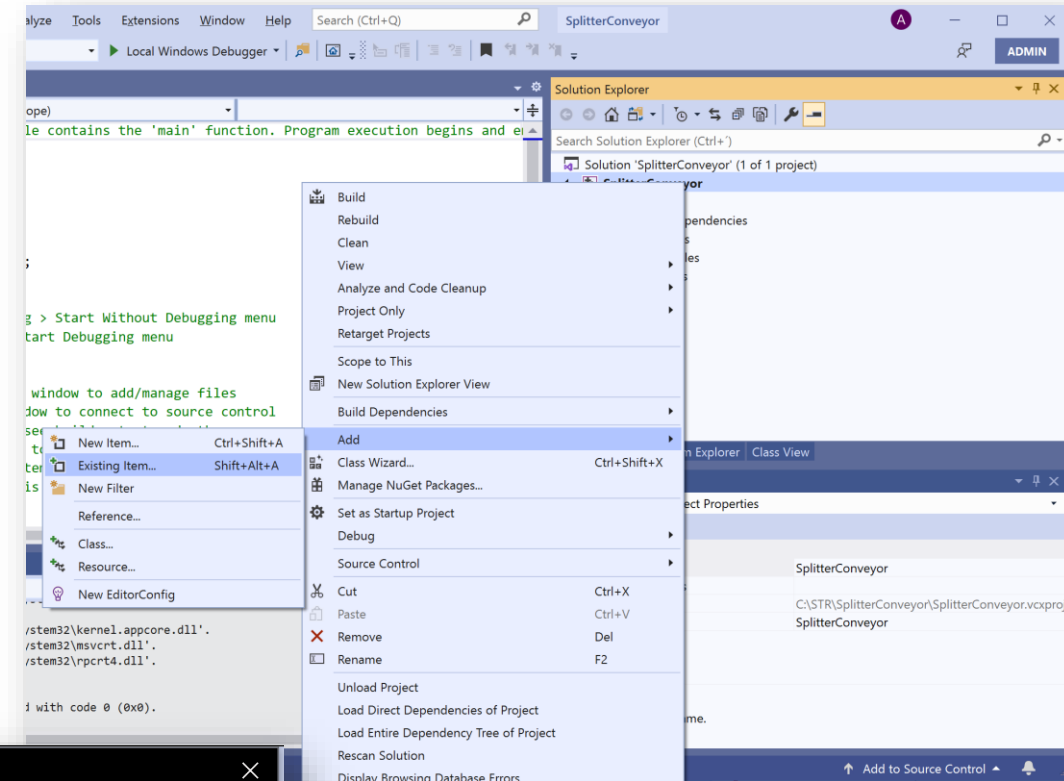
- ☐ Set the solution **platform to x64** bit
- ☐ Run the project to check if it is ok until now.



Creation of Visual studio console application project

Project Configuration for using the LineSplitter:

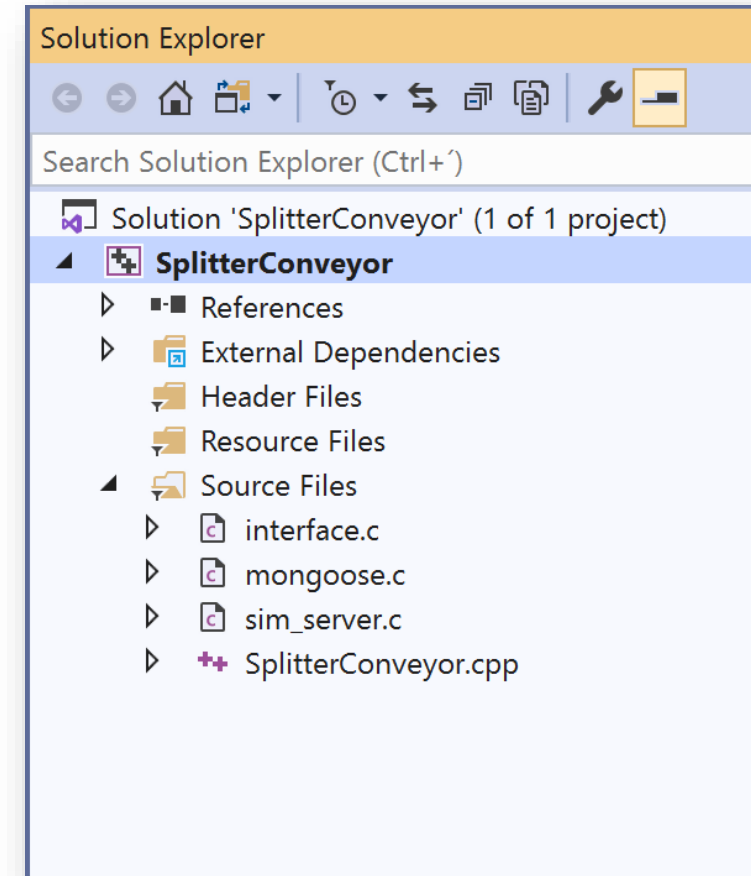
- ☐ Right-click on project and select **Add > Existing item**
- ☐ Select *.c files from **c:\str\source** folder and **add** to project



Creation of Visual studio console application project

Aspect of your solution explorer in Visual studio:

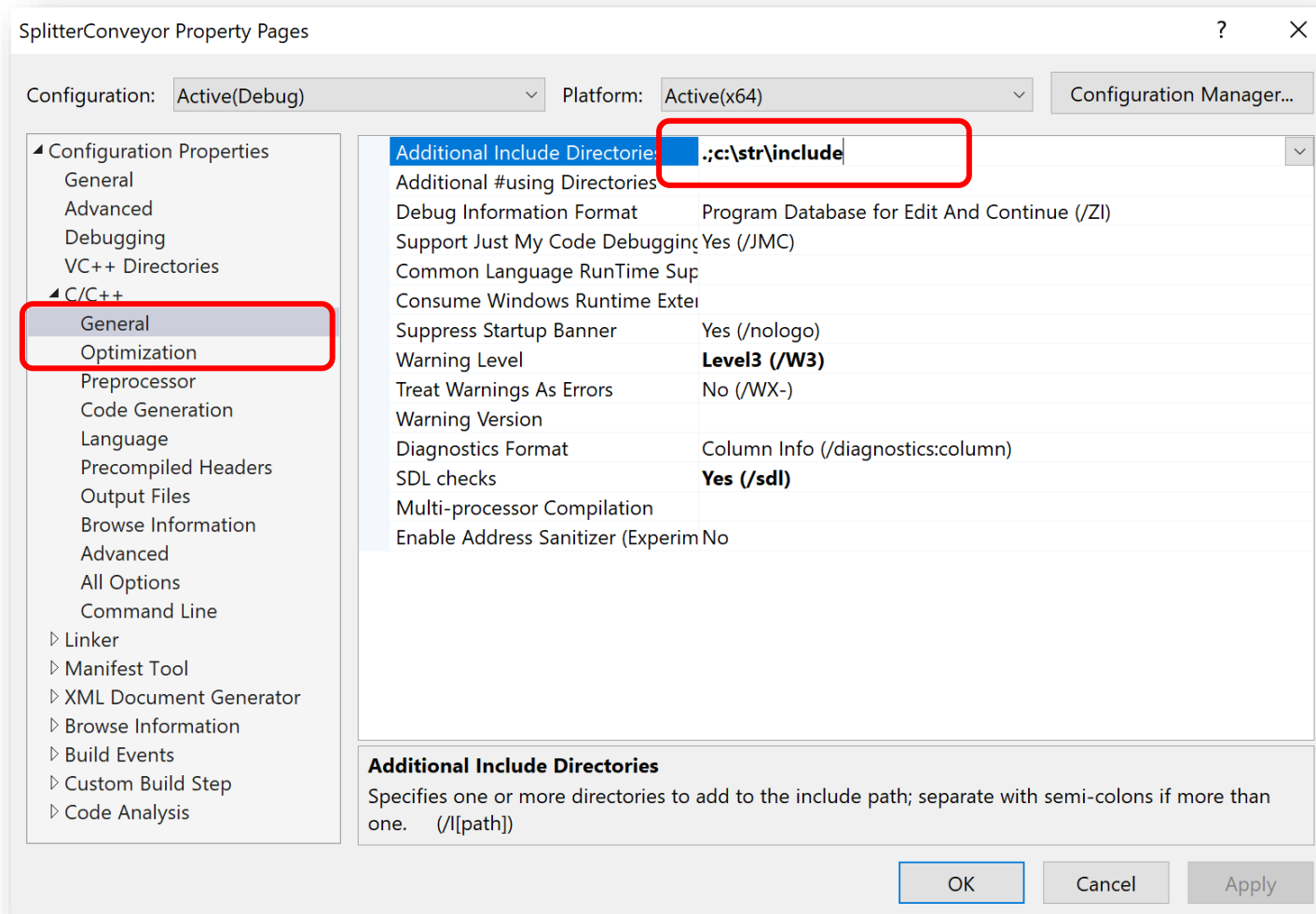
- ☐ **sim_server.c** - handles the interaction with the real kit or the simulator
- ☐ **interface.c** - allows switching between simulator and real kit.
- ☐ **mongoose.c** – embedded Web Server



Creation of Visual studio console application project



Configure include directories:

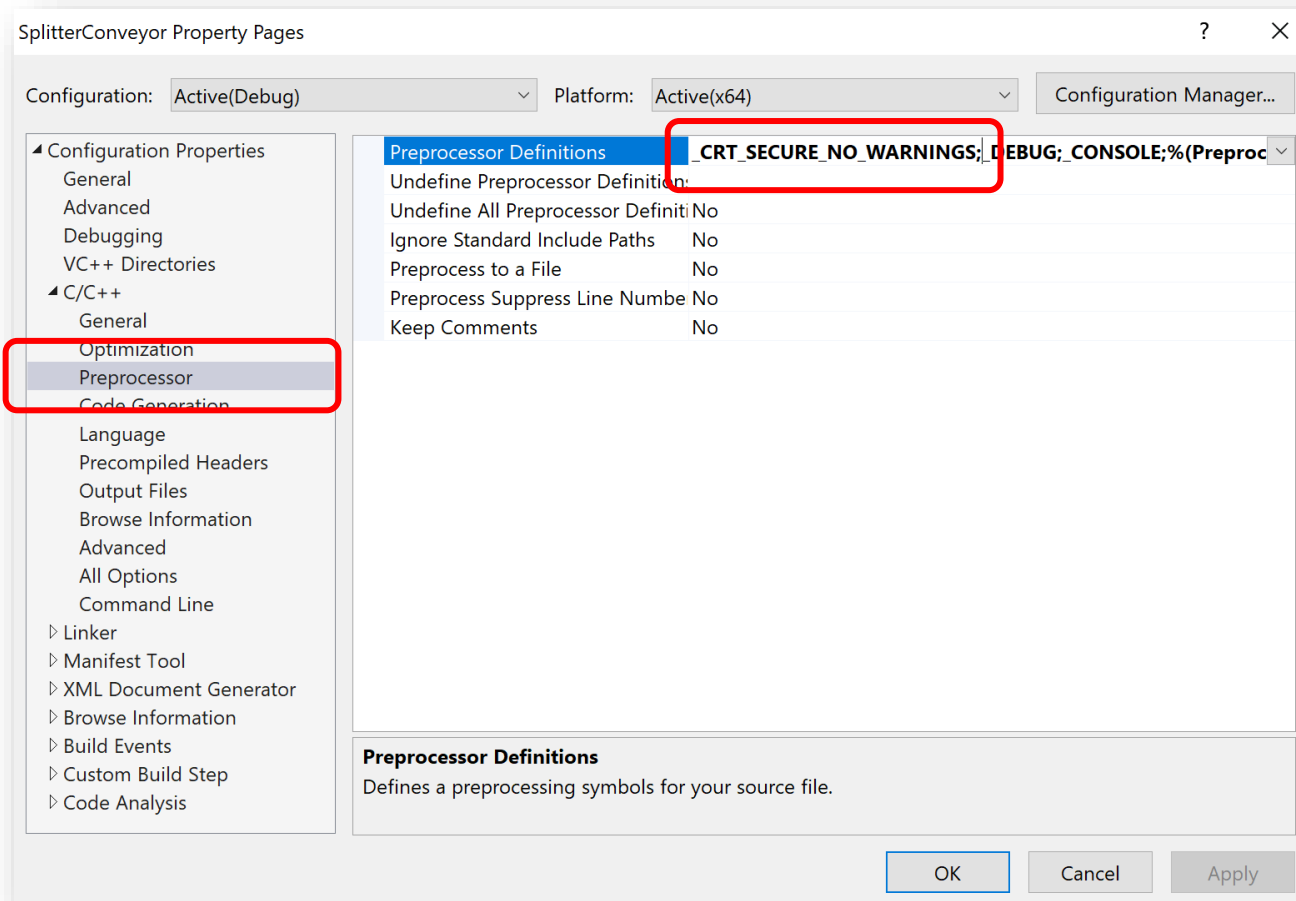


Copy past from here:
.;c:\str\include

Creation of Visual studio console application project



add “ CRT_SECURE_NO_WARNINGS” (in Project properties):



Copy past from here:

`_CRT_SECURE_NO_WARNINGS;`

Creation of Visual studio console application project



Try this example:

```
#include <stdio.h>
#include <conio.h>
#include <stdio.h>

extern "C" {
    // observe your project contents. We are mixing C files with cpp ones.
    // Therefore, inside cpp files, we need to tell which functions are written in C.
    // That is why we use extern "C" directive
    #include <interface.h>
}

int main()
{
    printf("Welcome to the Splitter Conveyor Application\n");
    printf("press key");

    createDigitalInput(0);
    createDigitalInput(1);
    createDigitalOutput(2);
    printf("\nPress a key to start");

    _getch();

    writeDigitalU8(2, 0x04);

    printf("\nPress a key to finish");

    _getch();

    closeChannels();
    return 0;
}
```

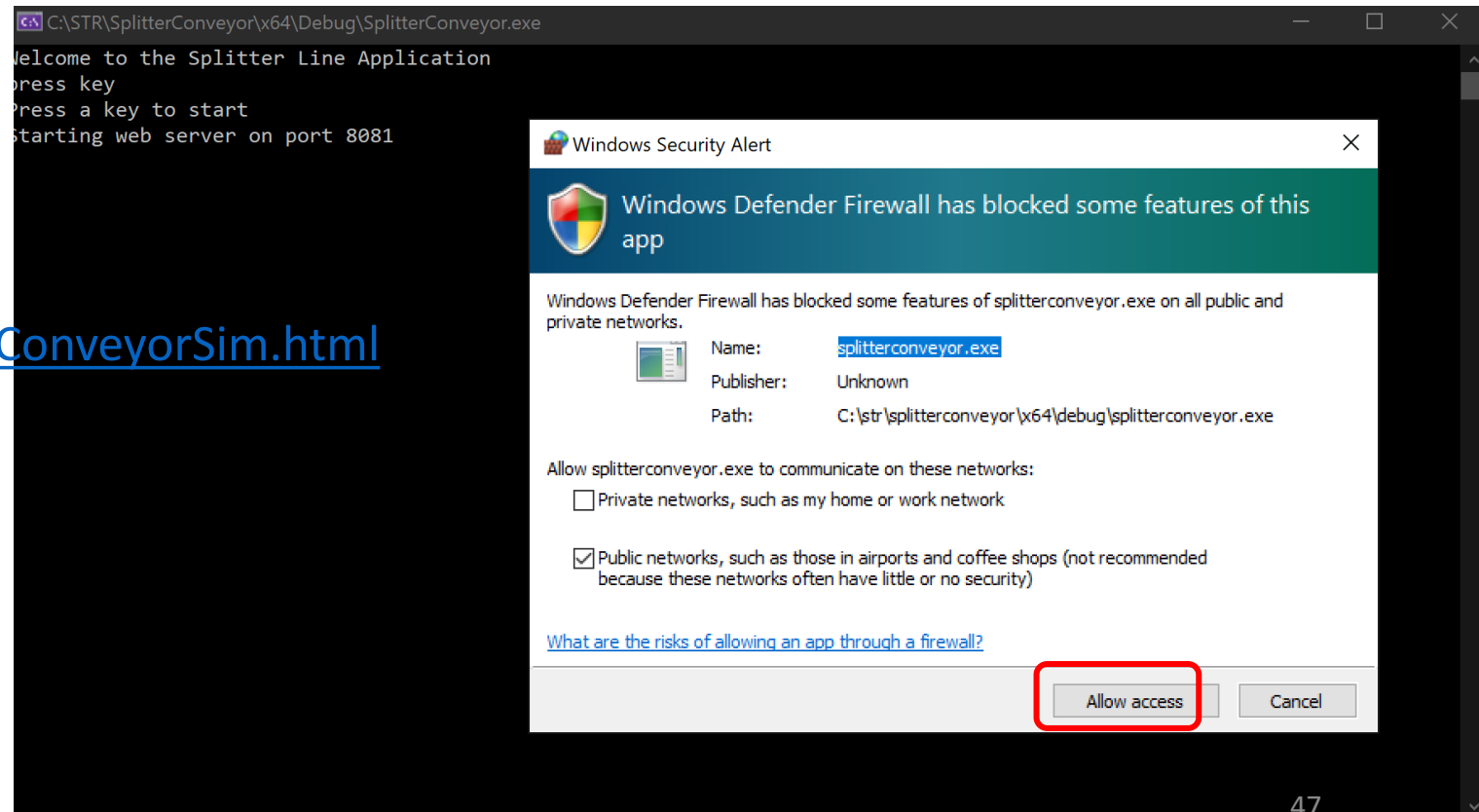
// Copy this code inside your source file
(probably the SplitterConveyor.cpp)

Creation of Visual studio console application project



Trying the application:

- ☐ Copy the file [SplitterConveyorSim.html](#) (available on clip) inside [c:\str\simulators](#) folder
- ☐ Run the application
- ☐ Press the allow access button
- ☐ Go into a browser and navigate to the address <http://localhost:8081/SplitterConveyorSim.html>



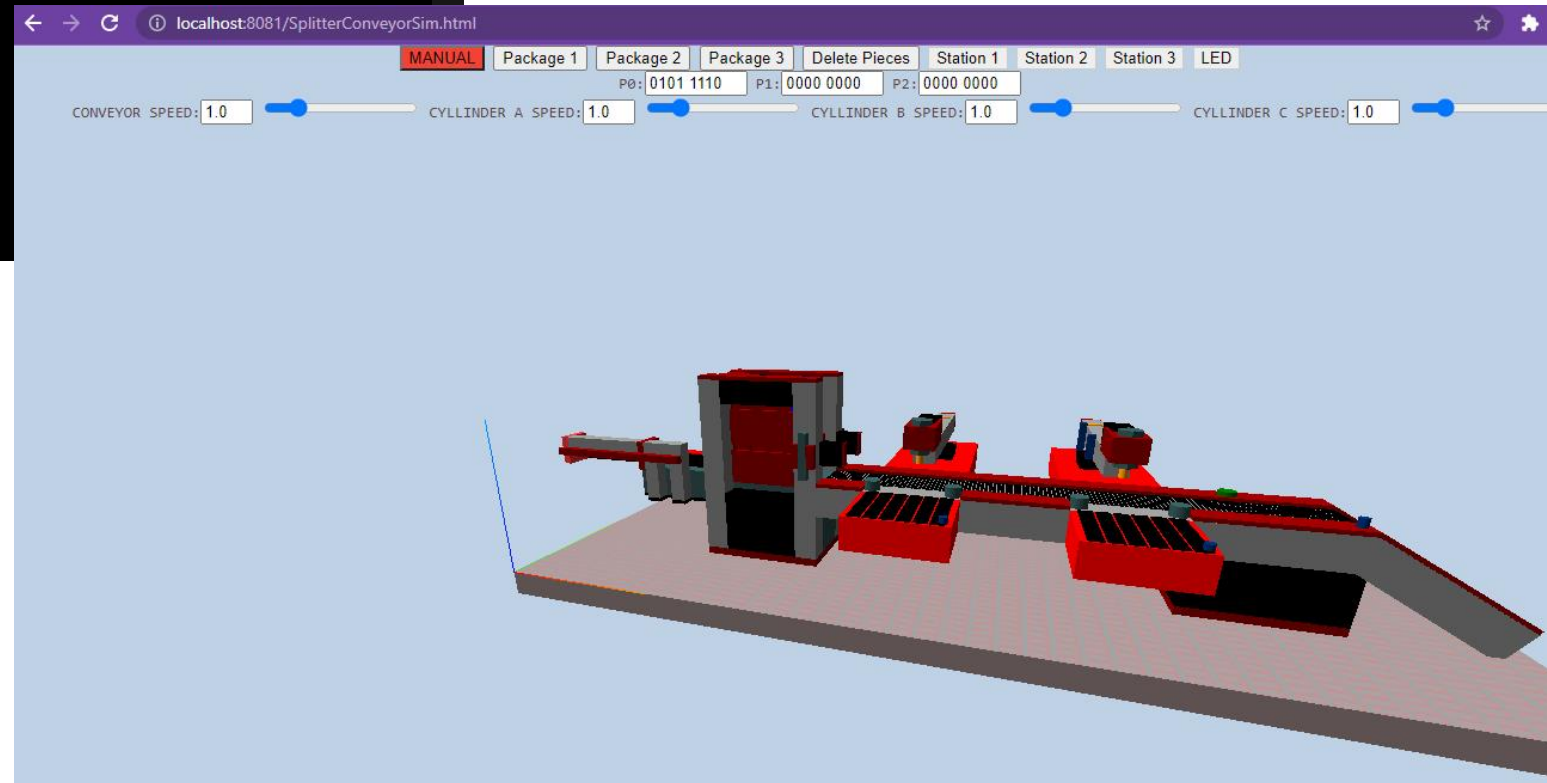
Creation of Visual studio console application project



<http://localhost:8081/SplitterConveyorSim.html>

Trying the application:

```
C:\STR\SplitterConveyor\x64\Debug\SplitterConveyor.exe
Welcome to the Splitter Line Application
press key
Press a key to start
Starting web server on port 8081
```



Case page not found:

❑ It means that simulator web server is working, but page **SplitterConveyorSim.html** is not found.

- Probably your libraries for **str** were copied into another folder, not the recommended one (**c:\str**)

❑ To solve, either

- Copy your libraries to the recommend folder or
- Modify macro **MG_HTTP_FOLDER** inside **interface.c** to the value corresponding to the location of the simulator folder, e.g “c:/str/simulators”

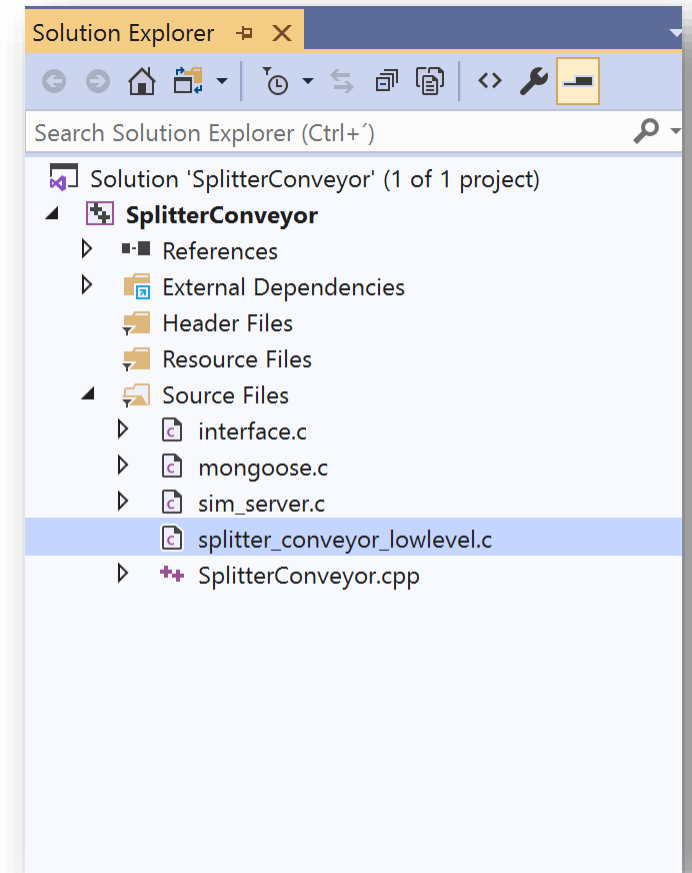


Creation of Visual studio console application project



Add new source file:

- ☐ Right click the project and select to add a new item (or press ctrl+shift+A).
- ☐ Add a new *.c file (not *.cpp).
- ☐ Name it splitter_conveyor_lowlevel.c



Creation of Visual studio console application project



Copy this code into splitter_conveyor_lowlevel.c:

```
#include <stdio.h>
#include <conio.h>
#include <stdint.h>

#include <interface.h>

void setBitValue(uint8_t *variable, int n_bit, int new_value_bit)
// given a byte value, set the n bit to value
{
    uint8_t mask_on = (uint8_t)(1 << n_bit);
    uint8_t mask_off = ~mask_on;
    if (new_value_bit) *variable |= mask_on;
    else               *variable &= mask_off;
}

int getBitValue(uint8_t value, uint8_t n_bit)
// given a byte value, returns the value of bit n
{
    return(value & (1 << n_bit));
}
```

Copy this code into splitter_conveyor_lowlevel.c:

```
int cylinder1_getPosition()
{
    int v = readDigitalU8(0);
    if (!getBitValue(v, 4))
        return 0;
    else if (!getBitValue(v, 3))
        return 1;
    return(-1);
}
```

```
void cylinder1_moveForward()
{
    uInt8 p = readDigitalU8(2);
    setBitValue(&p, 4, 1);
    setBitValue(&p, 3, 0);
    writeDigitalU8(2, p);
}

void cylinder1_moveBackward()
{
    uInt8 p = readDigitalU8(2);
    setBitValue(&p, 4, 0);
    setBitValue(&p, 3, 1);
    writeDigitalU8(2, p);
}

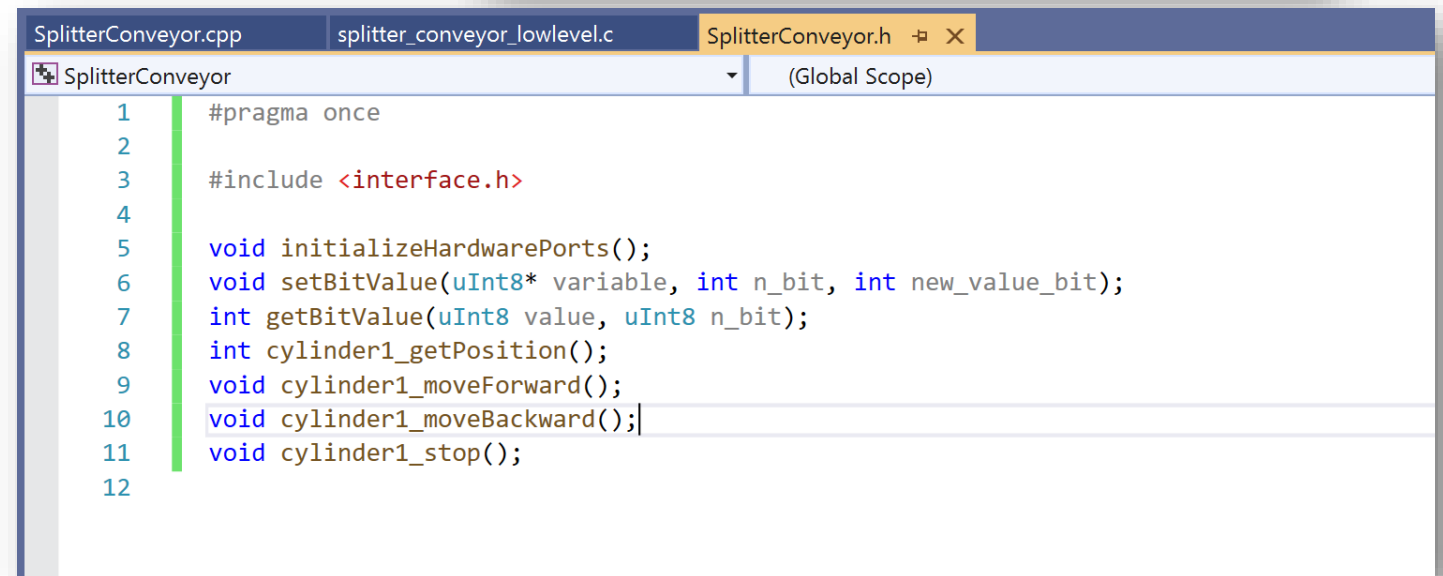
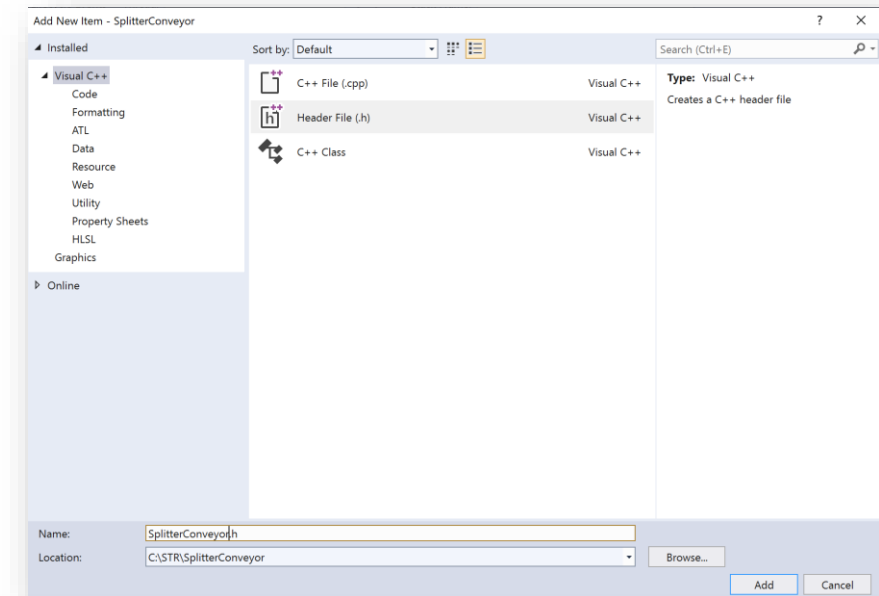
void cylinder1_stop()
{
    uInt8 p = readDigitalU8(2);
    setBitValue(&p, 4, 0);
    setBitValue(&p, 3, 0);
    writeDigitalU8(2, p);
}
```

Creation of Visual studio console application project



Add new file to project.h:

- ☐ Press “Shift+Ctrl+A”
- ☐ Select “Header File”
- ☐ Name it SplitterConveyor.h
- ☐ Add all the necessary signatures according to the picture at the right.



[illegible]

- **Test developed functions inside the main() function (complete code in the next slide)**

```
int main()
{
    printf("Welcome to the Splitter Conveyor Application\n");
    initializeHardwarePorts();
    int t = -1;

    printf("\n\nPress: q, a, z or 0 ");
    while (t != '0') {
        t = _getch();
        switch (t) {
            case 'q': cylinder1_moveForward(); break;
            case 'a': cylinder1_moveBackward(); break;
            case 'z': cylinder1_stop(); break;
        }
    }

    writeDigitalU8(2, 0x00);
    closeChannels();
    return 0;
}
```

Creation of Visual studio console application project



Edit the code:

We are Sorry! The code in the picture at the right must be typed.

... So that you can practice!



```
SplitterConveyor.cpp  x splitter_conveyor_lowlevel.c  SplitterConveyor.h
SplitterConveyor (Global Scope)
1  // SplitterConveyor.cpp : This file contains the 'main' function. Program execution begins and ends there.
2  //
3
4  #include <stdio.h>
5  #include <conio.h>
6  #include <stdio.h>
7
8
9  extern "C" {
10     // observe your project contents. We are mixing C files with cpp ones.
11     // Therefore, inside cpp files, we need to tell which functions are written in C.
12     // That is why we use extern "C" directive
13 }
14 #include <interface.h>
15 #include <SplitterConveyor.h>
16
17 int main()
18 {
19     printf("Welcome to the Splitter Conveyor Application\n");
20     initializeHardwarePorts();
21     int t = -1;
22
23     printf("\n\nPress: q, a, z or 0 ");
24     while (t != '0') {
25         t = _getch();
26         switch (t) {
27             case 'q': cylinder1_moveForward(); break;
28             case 'a': cylinder1_moveBackward(); break;
29             case 'z': cylinder1_stop(); break;
30         }
31     }
32
33     writeDigitalU8(2, 0x00);
34     closeChannels();
35     return 0;
36 }
37
38
```

NOVA

Actuators signature functions	keyboard
void cylinder1_moveForward()	q
void cylinder1_moveBackward()	a
void cylinder1_stop()	z
void conveyorMove()	
void conveyorstop()	
// complete the vocabulary	

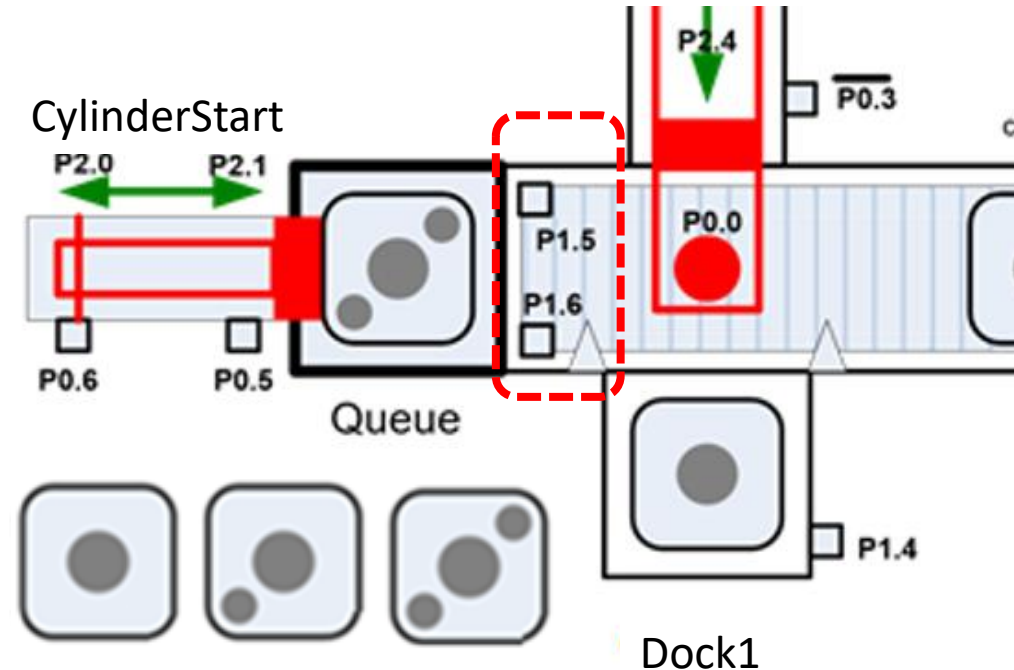
Complete the tables to achieve the following:

DAQ initialization	keyboard
void initializeHardwarePorts();	i

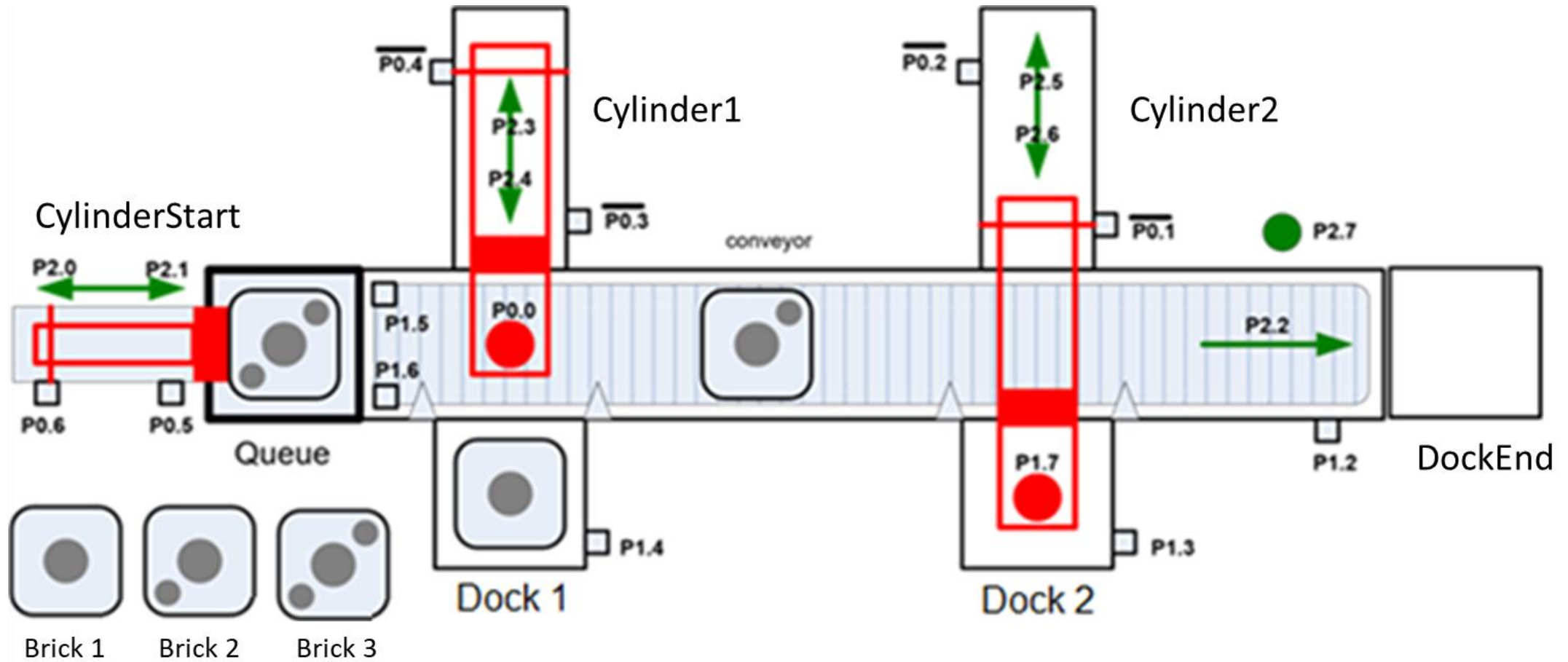
56

Functionality of “int getIdentificationSensors()”:

- ☐ This function must read port 1.
 - ☐ Then it must check if any magnetic sensor in P1.5 or in P1.6 is activated.
 - ☐ These sensors are never in the state activated (true value) at the same time.
 - ☐ If none sensors are in “true” state, the function returns 0.
 - ☐ If sensor in P1.5 is “true”, the function returns 1.
 - ☐ If sensor in P1.6 is “true”, the function returns 2.
- ☐ Remind: just the function is not enough to identify the brick destination. This will be handled at the Java Application level.



Introducing the kit “Splitter Conveyor”





KEEP UP THE
GOOD
WORK!